

General formatting requirements:

- Document must be single-spaced with double spaces between the paragraphs
- Pages must be numbered
- Must use a 12 point font

Note: When the project is ongoing, you will locate the *updates/change log*, *current status*, and *weekly summaries/meeting summaries* at the beginning of the document. When the project is complete, move these sections to appendices and delete the *current status* section

Updates/change log

[date]	[description of change]
10/03/2025	Update to client & project background
10/19/2025	Update to Competitive analysis and Heuristic Evaluation
10/26/2025	Lit Review & Project Plan Added
12/02/2025	User Field Research Plan
12/07/2025	UAM update for physicians
12/18/2025	UAM update for Pregnant women
12/21/2025	Final Fall Update
2/1/2026	Persona Update
2/8/2026	Affinity Map & Information Architecture Update
2/22/2026	Update to Information Architecture
3/12/2026	Low / Mid Fidelity Prototypes Added
4/15/2026	Usability testing protocol and structure added
4/21/2026	Usability Testing & Agentic Findings added
4/23/2026	Final Screens & Reasoning Added

This is also a good location for your weekly activity summaries and/or meeting summaries. Summaries should be dated (see below)

- Summary date: [MM/DD/YYYY]
- Summary notes: [summary note text here]

Current status

We are currently finishing up any last minute changes, and reformatting the doc.

Truepulse Health Capstone Project

Team members: Aiden Eichenour, Dinesh Bhathad, Fernando Luna, Sahil Deshpande

Date: April 23, 2026

[Updated Weekly Status Report](#)

Master of Science Program in Human-Computer Interaction

Department of Human-Centered Computing

Indiana University School of Informatics, Computing, and Engineering

Table of Contents

Updates/change log	1
Current status	2
Updated Weekly Status Report	3
Table of Contents.....	3
Executive summary.....	4
Introduction	5
Client Information.....	6
Project Information	7
Background	7
Research.....	10
Desk research (secondary research).....	10
Problem summary.....	10
Stakeholder Summary	10
Literature review	10
Autonomic Adaptations and Dysregulation During Pregnancy.....	11
Autonomic Dysregulation Across Pregnancy and Postpartum.....	11
Wearable Technology for Continuous ANS Monitoring	11
Challenges and Research Gaps	12
Implications for TruePulse	12
References	12
Field research (primary research).....	13
Competitive analysis	13
Evaluation	15
VitalScan ANS+	15
Whoop App	15
Empatica	19
Summary	20
Strengths and Weaknesses of Competitors	21
Opportunities, Threats, and Market Positioning	21
Novel Features to Consider	21
Essential Features for the Design	22

Other Discoveries Relevant to Future Designs	22
References	23
User research	23
Research Plan: Semi-Structured Interviews with Clinicians	23
1. Research Objective	23
2. Research Questions	24
3. Methodology.....	24
4. Participants	24
5. Interview Structure	26
6. Data Collection & Analysis	33
7. Ethical Considerations	33
8. Timeline (Estimated 3 Weeks)	33
Cognitive Walkthroughs	33
Cognitive Walkthrough Execution	34
Critical Design Implications.....	34
Analysis	35
Visual Analyses.....	35
Personas & User Journey Maps	35
Affinity Mapping	37
Link to Artifacts:.....	39
Outcome of Analysis: Functional Requirements	39
Application-level requirements	39
Functional Requirements.....	41
Describe Opportunity and Recommendations	42
Key Research Findings	43
Ideation	43
Information Architecture.....	44
Architectural Rationale	44
Task Analysis (Flow) - Pregnant Women	44
Initial Sketches/Low Fidelity	48
Mid Fidelity Prototypes	50
Onboarding and Sensor Setup	51
Home Dashboard	51

Profile and Sensor Management	52
Session History	52
Design System and Visual Language	52
High Fidelity Prototypes.....	53
Usability Testing.....	54
Usability Testing Methods	54
Participants	54
Protocols	54
Objective	54
Methodology	54
Tasks.....	54
Participant Profiles Tested	54
Venue	54
Usability Testing Results	54
1. Moderated Think Aloud	54
2. Test Documentation	54
2.1 Objective	54
2.2 Tasks.....	54
2.3 Participant Profiles.....	54
2.4 Venue	54
3. Findings	54
3.1 Summary of Results	54
3.2 What Surprised Us	54
3.3 Changes to the Prototype.....	54
Immediate Changes (High Priority).....	54
Future Enhancements (Lower Priority)	54
Agentic Testing	54
Objective	54
Tasks.....	54
Task 1: Log a Symptom Using My Journal	54
Task 2: Add a Recommended Product to Cart.....	54
Task 3: Interpret Personal Health Insights from Analytics	54
Participant Profiles.....	54

Venue.....	54
Findings.....	54
Summary of Results	54
What Surprised Us	54
1. The severity of the state persistence failure	54
2. The Shop crash was universal	54
3. Users in distress cannot compensate for friction	54
4. The 'Notes' field drew more attention than the custom symptom field	54
5. Analytics trust collapse was near-universal.....	54
Prototype Changes to Be Made.....	54
Final Prototype (key screens)	54
Changes Made After Usability Testing & Rationale.....	54
Overall Design Direction	54
References	54
Appendices	54
Pregnancy Monitoring Interview Summary (PW Participants)	54
A. Warm-up & Background.....	54
B. Understanding of Symptoms & Body Awareness.....	54
C. Current Tools & Monitoring Experience.....	54
D. Anxiety, Safety & Communication Needs	54
E. Technology Expectations & Feedback	54
F. Wrap-up	54
Clinician Interview Summary	54
Section A: Warm-up & Context	54
Section B: Workflow & Interpretation	54
Section C: Pain Points, Challenges & Gaps	54
Section D: Technology, Mobility & Communication.....	54
Section E: Wrap-up	54
Persona Drafts	54
Internal Evaluation.....	54
Cognitive Walkthrough Report (CWR).....	54
Task Description.....	54
Task Action Sequence	54

Interface / System Description	54
Cognitive Walkthrough	54
Test 1.....	54
Step 1: Begin Setup Sequence	54
Step 2: Setup Polar H10	54
Step 3: Begin Sit-Stand Sequence.....	54
Step 4: Follow Study Sequence.....	54
Step 5: Finish Test & View Results	54
Test 2.....	54
Step 1: Begin Setup Sequence	54
Step 2: Setup Polar H10	54
Step 3: Begin Sit-Stand Sequence.....	54
Step 4: Follow Study Sequence.....	54
Step 5: Finish Test & View Results	54
Potential Fixes for Identified Issues.....	54

Executive summary

Write this section last—it will feel repetitive because it is meant to summarize the entire document. Note that it is NOT meant to summarize the *project*, it is meant to summarize the *document*. In other words, the reader will find NO NEW INFORMATION here.

Introduction

Client Information

- Name of Client
 - Truepulse Health
 - Type of Business domain – Healthcare
 - Size of Org/organization's place in business domain – New Entry
 - Sujata Punait – spunait@tpulsehealth.com
 - Anil Malik – amalik@tpulsehealth.com

IU labs – The Launch Accelerator for Biosciences (IU LAB) is a first-of-its kind academic-industry initiative that will position Indiana as a National leader in bioscience discovery. Supported by a \$138 million Lilly Endowment Inc. Grant, IU LAB will harness Indiana's extensive resources in bioscience discovery, leading to significant advancements in human health.

- Names/email addresses of collaborators

IUB Build Lab Team — Collaborators

Fall Team —

- Sydney Shena — syshena@iu.edu
- Deborah Abioye — eabioye@iu.edu
- Matthew Widjaja — mawidj@iu.edu

Spring Team —

- Wonseok Choi — wc17@iu.edu
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- Trevor Peyton — trpeyton@iu.edu
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- Build Lab Leaders —

Bryce Himebaugh — bhimebau@iu.edu

Lewis, Gregory — lewigr@iu.edu

- Client goal for the project

- Interested in exploring how students might assist with the product development segment—specifically creating a portable prototype of our software solution. For this prototype, we envision using off-the-shelf physiological sensors that send signals to our app.
- Client objective for the design
 - A portable prototype with physiological sensors, connecting ubiquitous sensors to check ECG and respiration. With modular ability to add more sensors in the future. With visualization, and real-time information displayed.
 - Use of Ai to aid in recognition of activities, for noise identification. As well as the fully automated version of it to allow for less interaction to be corrected.
 - The language currently is Python but could change with the new version due to it being portable.
 - Simple UI with a clear way for users to make sense of data. With Easy errors to showcase the error in sensor placement or an error.
- Constraints (such as branding requirements, look and feel, device constraints, etc.)
 - There is a constraint on Ansense, so they would need a new application similar.

Project Information

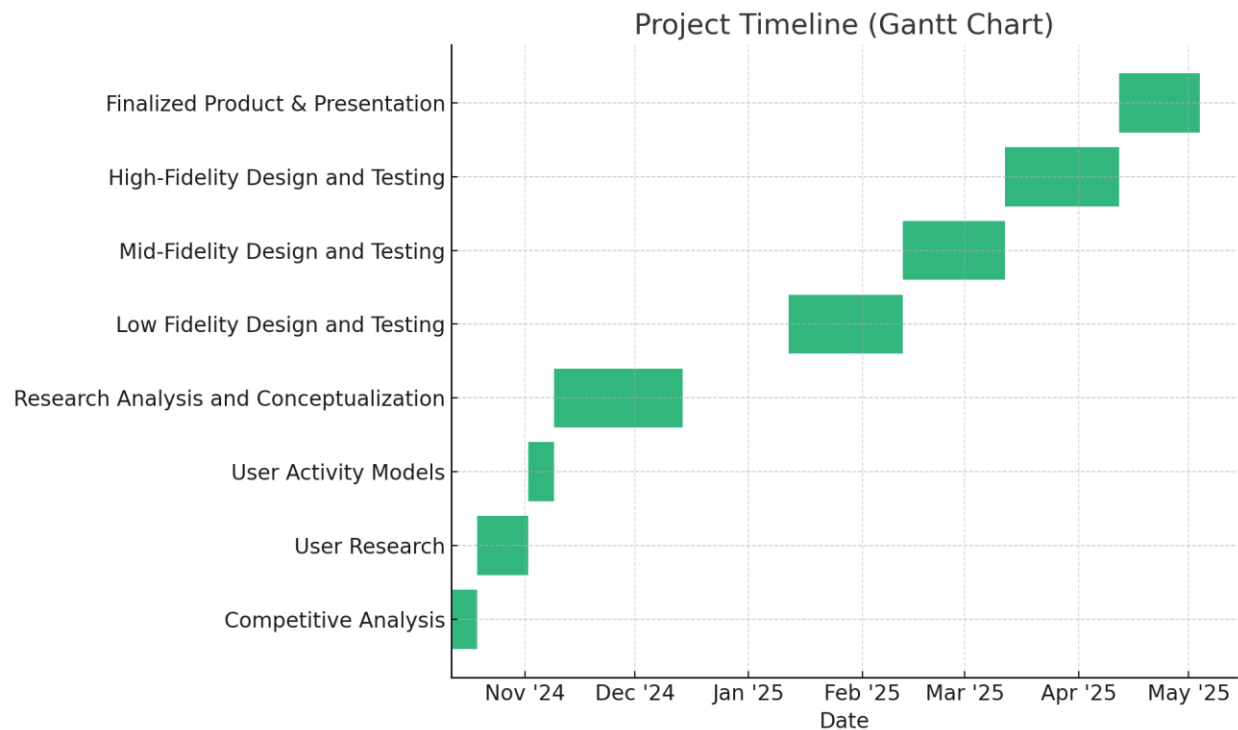
Background

Description of the field

- The product tracks data related to the Autonomic Nervous system; the main functions are stress response and recovery and rest. Sympathetic, ramps the body up aka fight or flight response. Parasympathetic calms the body down, bringing a sense of calmness throughout the body. There can be an ANS imbalance when pregnant, which puts the body under stress with constant changes throughout the pregnancy. This can lead to Autonomic dysfunction. The prototype aims to provide data within a clinical sense to enhance the ability to track the data.
- This will allow hospitals and clinics to purchase this to add to their system. Initially, being a B2B solution. Within the outpatient lab, and outpatient clinics. Small potential for a B2C solution further down the line. Outside of the work we'd be doing.
- The product aims to provide a way to monitor autonomic dysfunction in pre-eclamptic women. To empower a new approach to maternal care, allowing more information over ANS.

Project Management Tools – Teams, Outlook

Project plan (activities, milestones, and dates)



Task (Course Deliverables)	Timeline
Competitive Analysis	October 12 – 19
User Research	October 20 – November 02
User Activity Models	November 03 – November 09
Research Analysis and Conceptualization	November 10 – December 14
Winter Break	
Low Fidelity Design and Testing	January 12 – February 12
Mid-Fidelity Design and Testing	February 13 – March 12
High-Fidelity Design and Testing	March 13 – April 12
Finalized product & Presentation	April 13 – May 04

- **Partner Resources:** The Truepulse Health team will provide access to relevant documentation and hardware specifications for compatible physiological sensors. Any required data or software access for prototype testing will be coordinated through Sujata Punait. Sujata will also share periodic feedback on progress and design during review meetings.
- **Communication Methods/Frequency:** The team will use Microsoft Teams as the central platform for communication, scheduling, and file sharing. Email serves as the primary for formal updates and deliverable sharing with the Industry Partner. Internal

communications will happen through the team's shared Teams channel, with meeting notes being shared and stored in the project folder.

- Meeting Frequency:
 - Internal Team Meetings: Weekly on Wednesdays to discuss progress, assign tasks, and plan for upcoming milestones.
 - Partner Meetings: Bi-weekly meetings on Fridays, adjusting if needed due to conflicting events.
 - Potential Check Ins: Additional short syncs may be conducted as needed to address design or technical questions. Conducted through Email, or Teams depending on group.
- File Sharing Plan: Project files will be maintained in a shared Teams Folder with organized subfolders for deliverables, research, design iterations, and meeting summaries. Backup copies of major deliverables will also be stored on OneDrive to ensure version control and data security.
- Spokespeople for the project:
 - Aiden Eichenour – Designer & Leading Contact
 - Dinesh Bhathad – Researcher
 - Fernando Luna – Researcher & Leading Contact
 - Sahil Deshpande – Designer
- Challenges (you can update this section as challenges are resolved, and new challenges arise)
 - The earlier concern regarding the need for IRB approval has been resolved, as it is no longer required for this project.
 - Can't directly obtain the current demo for AnSense, that has been implemented. May change due to Sujata getting clearance.
 - The IU policy that isn't allowing Sujata to share more details
 - What we can show for capstone going further is unknown for demonstration.
- Tensions

- The timeline of the IUB Build Lab team, with them ending up their portion this December. A new team will be assigned next semester to continue what the current team is unable to finish.
- Delay in receiving Documentation and Resources, for the project.

Research

Desk research (secondary research)

Problem summary

Maternal health during pregnancy and the postpartum period is tightly coupled to the functioning of the autonomic nervous system (ANS), which governs cardiovascular, respiratory, and stress-related responses. Complications such as preeclampsia, preterm birth, and postpartum hypertension are often preceded by detectable autonomic imbalances — including elevated sympathetic tone, reduced parasympathetic activity, and impaired baroreflex sensitivity — that manifest days or weeks before clinical symptoms appear.

Despite this, current standard-of-care assessments (intermittent blood pressure checks, laboratory biomarkers, and symptom-based evaluations) are episodic and cannot capture the continuous dynamics of ANS function. There is no widely adopted, hardware-agnostic, real-time ANS monitoring solution designed specifically for the perinatal period that integrates multimodal physiological signals into a clinically actionable platform.

TruePulse Health is building a portable, software-based ANS monitoring prototype targeting this gap — connecting off-the-shelf ECG and respiration sensors to an intelligent application capable of real-time visualization, AI-assisted signal interpretation, and early complication alerting.

Stakeholder Summary

Sujata Punait — Primary client contact and product vision holder. Sujata drives TruePulse's clinical and scientific strategy, ensuring the platform addresses real unmet needs in maternal care. As a stakeholder, she is invested in a prototype that demonstrates clinical plausibility, sensor reliability, and a clear path toward hospital integration.

Anil Malik — Technical co-founder overseeing the software and hardware architecture of the platform. Anil's priorities center on building a modular, hardware-agnostic system — initially in Python — that can be adapted for portability and expanded with additional sensor types over time. He is the key decision-maker on technical feasibility and implementation constraints.

Literature review

Maternal health during pregnancy and postpartum is deeply intertwined with the functioning of the autonomic nervous system (ANS), which regulates cardiovascular, respiratory, and stress-related physiological responses. Across the perinatal period, fluctuations in heart rate variability (HRV), blood pressure variability (BPV), and other autonomic markers provide early insight into maternal well-being and the onset of complications such as preeclampsia or preterm birth. However, traditional clinical assessments such as intermittent blood pressure checks, laboratory biomarkers, and symptom-based evaluations are episodic and fail to capture the continuous dynamics of autonomic function. Recent research integrating wearable technology, perinatal physiology, and cardiovascular science underscores the potential of continuous ANS monitoring to transform maternal care from reactive diagnosis to proactive prevention.

Autonomic Adaptations and Dysregulation During Pregnancy

Normal pregnancy involves extensive autonomic adjustments to support fetal development. Reyes et al. (2018) demonstrated that sympathetic nerve activity increases by approximately 50–150% during pregnancy, which helps maintain blood volume and cardiac output while vasodilatory and hormonal mechanisms preserve hemodynamic balance. In hypertensive or preeclamptic pregnancies, this equilibrium is disrupted; sympathetic drive remains excessive, parasympathetic modulation weakens, and vascular resistance rises leading to symptoms such as persistent headaches, dizziness, and elevated heart rate. For expectant mothers, this imbalance can cause heightened anxiety, fatigue, and physical discomfort, directly affecting daily functioning and overall well-being. Similarly, Yousif et al. (2019) synthesized data from 26 studies and reported that over 90% of women with preeclampsia exhibited autonomic dysfunction characterized by elevated sympathetic tone, reduced parasympathetic activity, and impaired baroreflex sensitivity. These findings indicate that autonomic imbalance precedes clinical symptoms, suggesting that continuous ANS monitoring could enable early detection of maternal complications.

Autonomic Dysregulation Across Pregnancy and Postpartum

Although preeclampsia is often identified in late pregnancy, autonomic instability persists after delivery. Van Wingerden et al. (2024) observed that women with preeclampsia exhibited significantly lower HRV and higher BPV during the first week postpartum than normotensive controls, reflecting sustained sympathetic dominance and decreased cardiovascular flexibility. Persistent dysregulation may explain why women with prior preeclampsia experience heightened long-term risk of hypertension, stroke, and cardiovascular disease. These results highlight a gap in clinical follow-up, where postpartum physiological recovery is rarely monitored despite continued vulnerability.

Wearable Technology for Continuous ANS Monitoring

Emerging studies demonstrate that consumer wearables can provide reliable, longitudinal ANS data. Rowan et al. (2022) validated the WHOOP strap for continuous tracking of maternal HRV and resting heart rate across pregnancy and postpartum, capturing expected declines in HRV until approximately 33 weeks of gestation and its rebound before delivery. Claydon et al. (2024) extended this work by showing that wearable-derived sleep and activity data reflect concurrent autonomic and behavioral changes across the perinatal period, providing a multidimensional view of maternal adaptation. Building on these findings, Jasinski et al. (2024) identified HRV “inflection points” that corresponded more closely with time-to-delivery than gestational age, allowing preterm birth prediction up to seven weeks in advance. Collectively, these studies validate that noninvasive HRV tracking through wrist- or chest-based sensors offers a feasible and scalable method for continuous maternal ANS monitoring.

Challenges and Research Gaps

Despite promising advances, existing studies rely heavily on proprietary hardware (e.g., WHOOP), limiting interoperability and integration into hospital systems. Most focus on single-modality metrics such as HRV without combining multimodal physiological signals like respiration, oxygen saturation, or electrocardiography (ECG). Furthermore, although predictive modeling has emerged, current systems cannot translate autonomic patterns into actionable, context-aware insights for clinicians and patients. The literature, therefore, points to the need for hardware-agnostic, integrative, and clinically compatible ANS platforms that unify diverse bio signals, analyze them in real time, and provide personalized feedback.

Implications for TruePulse

TruePulse’s mission is to develop a hardware-agnostic, real-time ANS monitoring platform for maternal health directly addresses these clinical and technological gaps. By integrating ECG, respiration, and pulse-oximetry data into a unified system, TruePulse can deliver continuous maternal monitoring across prenatal, intrapartum, and postpartum stages. Its analytics can quantify HRV, BPV, and respiration variability to detect early signs of sympathetic overactivity or parasympathetic withdrawal linked to preeclampsia, preterm birth, or postpartum hypertension. Through customizable alerts, AI-assisted pattern recognition, and intuitive dashboards, TruePulse could empower clinicians and mothers alike, thus transforming maternal health from episodic assessment to continuous, predictive, and preventive care. In doing so, TruePulse bridges consumer wearables and clinical decision support, aligning directly with the growing body of evidence advocating for continuous ANS-based monitoring in maternal care.

References

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Rowan, S. P., Lilly, C. L., Claydon, E. A., & Capodilupo, E. R. (2022). Monitoring one heart to help two: Heart rate variability and resting heart rate using wearable technology in active women across the perinatal period. *BMC Pregnancy and Childbirth*, 22, 887. <https://doi.org/10.1186/s12884-022-05183-z>

Van Wingerden, A.-S., Katsidoniotaki, M., Haghighi, N., Almonte, C., Martinez, H. W., Valdes, E., ... Miller, E. C. (2024). Postpartum blood pressure variability and heart rate variability in preeclampsia. *Hypertension*, 81, 2510-2519. <https://doi.org/10.1161/HYPERTENSIONAHA.124.23321>

Yousif, D., Bellos, I., Penzlin, A. I., Hijazi, M. M., Illigens, B. M.-W., Pinter, A., & Siepmann, T. (2019). Autonomic dysfunction in preeclampsia: A systematic review. *Frontiers in Neurology*, 10, 816. <https://doi.org/10.3389/fneur.2019.00816>

Field research (primary research)

Competitive analysis

1. VitalScan ANS+ is a diagnostic system that evaluates autonomic nervous system function using HRV, ECG, PPG, and sudomotor analysis. Mainly used within clinics and lab settings.
2. WHOOP App monitors heart rate variability (HRV), resting heart rate, sleep architecture, and recovery, all key indicators of autonomic nervous system (ANS) activity. It has been used in maternal studies to track physiological stress and recovery throughout pregnancy and postpartum, making it a relevant proxy for developing ANS monitoring tools for new mothers.
3. Empatica Care is a remote patient monitoring platform that allows clinicians to monitor real-time physiological data from multiple individuals through a dashboard. It aligns with the project goal of ANS monitoring for maternal health by showing how complex biometric data (HR, HRV, temperature, respiration) can be visualized and managed across large populations.

Competitor	Cost	Affordability	Availability	Tracking range/Tech Inclusion	Types of Users (novice, expert, other comparisons)
VitalScan ANS+	Varies est. \$12-\$40k	Low – High cost for commercial use in clinics and hospitals.	Moderate – devices sold commercially , not to the mass public.	Computer Analysis: Yes; HRV: Yes; Beat to Beat Blood Pressure: Yes; Pulse Wave Velocity: Yes; 3 Lead ECG: Yes;	Clinicians, Doctors, Nurses
Whoop App	\$239 for device + \$30/month subscription	Moderate - affordable for health-conscious users but expensive for large-scale maternal adoption	High - available globally online and in retail stores	Continuous HRV: Yes; Sleep analytics: Yes; Temperature monitoring: Yes; EDA/GSR: No; Raw ECG access: No; Mobile app feedback: Yes; Pregnancy validation: Yes;	Athletes, wellness enthusiasts, pregnant and postpartum women seeking physiological insights
Empatica App	For one EmbracePlus device + 3-year data	Low – High-cost relative to consumer wearables;	High – Available globally via Empatica	Heart rate & HRV monitoring: Yes;	Clinicians, caregivers, researchers,

	access + Care Portal: US \$1,080 for the 3-year bundle in the “Academic & Basic Research” plan.	targeted at research/institution setting rather than mass-consumer.	store; requires compatible smartphone for Care Lab App.	Temperature monitoring: Yes; Bluetooth connectivity: Yes; User dashboard: Yes; Real-time vitals visualization: Yes; Historical trend view: Yes; Multi-user management : Yes; Cloud storage: Yes;	and high-risk patients
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Evaluation

VitalScan ANS+

Heuristic	Finding	Severity	Notes
Recognition vs Recall / User Control & Freedom	Complex Test Protocol	3	Diverse number of tests possible to do, with no clear key areas to navigate. Requires prior background with tests
Aesthetic & Minimalist Design / Visibility of System Status	Diagnostic Breadth	Positive	The key graphics showcasing the readings in real time allow for a good way to monitor various readings.
Help & Documentation / Match Between System & Real World	Output Opaque to Non-Specialists	2	Needs implementation of basic level interpretation for non-specialist use. With the inclusion of key words to allow for ease of interpretation.

Whoop App

Heuristic	Screens	Observation / Strengths (Maternal Context)
Visibility of System Status	Sleep Performance, Recovery, Stress Monitor	The app's visual feedback (color-coded recovery rings, HRV trends, sleep graphs) provides instant insight into physiological status, which could help pregnant or postpartum users track stress, fatigue, and recovery. However, physiological metrics such as HRV and strain fluctuate naturally during pregnancy; without contextual adaptation, users may misinterpret "low recovery" as a health issue.
Match Between System & Real World	Journal, Sleep, Recovery	Uses relatable wellness terms such as "Sleep Debt," "Recovery," and "Performance," aligning with daily maternal experiences of rest and fatigue. The Journal allows behavior logging (e.g., caffeine, alcohol), relevant to prenatal tracking. However, it lacks pregnancy-specific factors such as nausea, fetal movement, or contractions.
User Control and Freedom	Journal, Activity Log, Alarm Settings	Simple toggles and quick entries make daily interaction easy, even for users experiencing fatigue or limited mobility. Undo and edit functions provide flexibility. Voice input or guided journaling could further assist during late

		pregnancy or postpartum recovery when physical effort is limited
Consistency and Standards	All Screens	The app maintains consistent iconography, layout, and interaction patterns, reducing cognitive load. This is beneficial for pregnant users managing hormonal or cognitive fatigue. Technical terms like HRV or RMSSD could still be supplemented with plain-language explanations to improve accessibility
Error Prevention	Journal, Recovery Impact Analysis	Pre-set sliders and limited free-text input reduce entry errors—useful when dexterity or concentration may fluctuate during pregnancy. The system’s automation for activity detection minimizes mistakes. However, emotional misinterpretation of data (e.g., low recovery perceived as danger) may occur without contextual messaging.
Recognition Rather Than Recall	Recovery Dashboard, Sleep Overview	Visual gauges and color-coded trends minimize memory effort. This supports users who may experience “pregnancy brain” or postpartum cognitive fog. Adding short textual summaries like “Your HRV is stable this week” could

		further enhance understanding.
Flexibility and Efficiency of Use	Week Planner, Recovery Impact Analysis	Modular dashboards enable both quick glances and detailed exploration. This duality supports users with variable daily energy levels during pregnancy or postpartum. Allowing customized maternal widgets (e.g., hydration, contractions, sleep quality) could make it more relevant.
Aesthetic and Minimalist Design	All Screens	The dark theme and simple visual hierarchy reduce visual fatigue and fit nighttime use, valuable for new mothers with irregular sleep patterns. Data-dense screens like Recovery Impact Analysis might still overwhelm; a simplified “maternal mode” could present essential metrics more gently.
Help Users Recognize, Diagnose, and Recover from Errors	Sync Status, Journal	Stable syncing and auto-save features prevent frustration or data loss. If readings fluctuate due to pregnancy-related physiology (e.g., elevated HR, edema affecting PPG accuracy), the app currently lacks contextual reassurance. A short notification explaining normal physiological variance could improve trust.
Help and Documentation	Learn Tab, Journal	The “Learn” tab offers wellness education that

		could be extended to include prenatal and postpartum topics such as fatigue, HRV changes, and stress management. This would support self-understanding and reduce anxiety caused by misinterpreting normal physiological changes
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Empatica

Heuristic	Screen / Instance	Observation / Strengths	Issue / UX Gap
Visibility of System Status	Patient App – Home Screen (heart rate, temp, system status)	Displays real-time wearing time, heart rate, skin temperature, battery level, connectivity, and sync status. Indicators are clean and glanceable.	The status indicators are technically descriptive but lack interpretive cues for patients (e.g., what does “31.9 °C” mean for me? Is that good or concerning?). For pregnant users, this can increase anxiety or confusion.
Match Between System and Real World	Clinician Dashboard – Site View (multi-patient list)	Uses medical terminology that aligns with hospital workflows (e.g., HR, BPM, skin temp, sync time). Clear timestamping supports clinical reasoning.	Terminology is overly technical for non-specialist caregivers or maternal health workers. No quick legends, tooltips, or plain-language equivalents to help novice interpreters.
Recognition Rather than Recall	Clinician Dashboard – Trend View (graph visualization)	Clear graphs showing median, quartile bands, and daily trends make	Clinicians must mentally map which metric they’re observing (e.g., temperature vs heart

		longitudinal data easy to scan.	rate), as color coding is subtle and legends are light. For multi-patient monitoring, this adds cognitive friction.
Flexibility & Efficiency of Use	Clinician Dashboard – Site Navigation Panel	Supports rapid switching between wards/sites (e.g., Oncology, ICU, Post-ICU) and bulk viewing.	No clear alert prioritization or filtering — clinicians monitoring dozens of patients must scan all rows manually, which slows triage.
Aesthetic and Minimalist Design	Patient App – Real-time vitals view	The UI is clean, minimalist, and focused on only 3–4 essential metrics. Great for low cognitive load.	Lack of contextual reassurance for abnormal or fluctuating readings (e.g., “slightly above baseline but within normal range”) could cause uncertainty or panic, especially in pregnancy contexts.
Help Users Recognize, Diagnose, and Recover from Errors	System Status section – Patient App	Shows connectivity and sync status, which is transparent and helpful.	If the band disconnects or sync fails, messaging is technical rather than guiding (“E4 Wristband disconnected”), lacking actionable advice like “Reattach band” or “Check Bluetooth.”

Summary

We conducted a focused competitive analysis to guide our UX design decisions for maternal health monitoring. Our goal was to understand how existing systems communicate

physiological data to both patients and clinicians - balancing accuracy, interpretability, and emotional reassurance.

We selected three competitors to reflect this dual-user lens: VitalScan (clinician-facing diagnostic tool), WHOOP Strap 4.0 (consumer wellness wearable), and Empatica Care (hybrid clinician–patient platform). Each was analyzed for cost, feature set, user types, and usability using Nielsen Norman heuristics, focusing on visibility, interpretability, error prevention, and clarity. This provides a clear foundation for identifying design opportunities where TruePulse can stand out.

Strengths and Weaknesses of Competitors

VitalScan ANS+ provides comprehensive physiological analysis (HRV, ECG, PPG, BP, sudomotor activity) suited for clinical diagnostics. Its strength lies in diagnostic accuracy and the breadth of data visualization for specialists. However, this tool is not intended for non-specialists; its interface is dense, lacks guidance, and requires prior medical knowledge.

The WHOOP App excels in accessibility and engagement. It visualizes HRV, recovery, and sleep data with intuitive, color-coded displays. Its terminology and design align with user lifestyles, making it well-suited for wellness and pregnancy tracking. Weaknesses include a lack of pregnancy-specific context and the potential for data misinterpretation without medical framing.

Empatica Care demonstrates strong scalability and professional credibility, enabling multi-user monitoring and longitudinal tracking for clinical or research use. Its minimal design reduces cognitive load, but the system uses technical medical language that can alienate lay users. Lack of interpretive cues and alert prioritization limits real-time decision support.

Opportunities, Threats, and Market Positioning

There is a clear opportunity for a mid-tier, hybrid solution bridging consumer wearables and clinical systems. Current tools either target professional diagnostic use (VitalScan, Empatica) or general wellness (WHOOP), leaving a gap for pregnancy-specific autonomic monitoring that interprets data contextually. The threat lies in high competition from established brands and user distrust caused by inaccurate or decontextualized feedback. Market positioning for a new entrant like Truepulse can emphasize clinical reliability with consumer usability, an affordable, interpretable, AI-assisted maternal ANS tracker that adapts to real physiological changes rather than static baselines.

Novel Features to Consider

- Introduce context-adaptive interpretation, AI models that recognize pregnancy-related physiological variance and provide reassurance or alerts accordingly.
- Add maternal-specific data layers (hydration, contractions, fetal movement) and integrate guided journaling or voice-based logging for accessibility during fatigue.
- A hybrid dashboard combining clinician and self-use views can merge precision with simplicity.
- Contextual education modules similar to WHOOP's "Learn" tab, but medically verified, could reduce misinterpretation and anxiety.

Essential Features for the Design

- Core features should include real-time ANS visualization (HRV, ECG, respiration) with automated artifact detection and sensor error feedback.
- A simple, modular UI enabling users to switch between summary and detailed views is critical.
- Error prevention through automation and plain-language explanations should be built in.
- Data syncing, cloud storage, and secure sharing with clinicians are essential for clinical trust and regulatory alignment.

Other Discoveries Relevant to Future Designs

- User confidence depends on contextual explanation rather than raw metrics; numerical accuracy alone is insufficient for comprehension.
- Terminology standardization across user types (clinicians, pregnant women, caregivers) is necessary for cross-context usability.
- Design must support emotional reassurance during physiological fluctuations and progressive disclosure of complex data to reduce cognitive load.
- Cross-platform interoperability, allowing connection with off-the-shelf sensors, will strengthen scalability and long-term adoption.
- Several studies have validated WHOOP's use in prenatal, pregnancy, and postpartum monitoring. Rowan et al. (2022) showed that their HRV and resting heart rate data effectively track physiological changes across the perinatal period. Jasinski et al. (2024) identified wearable-derived maternal HRV as a biomarker for preterm birth, while additional research (Rowan et al., 2024) confirmed its accuracy in detecting sleep and activity shifts. These findings support WHOOP as a validated wearable for maternal health insights.

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User research

Research Plan: Semi-Structured Interviews with Clinicians

1. Research Objective

- To identify pain points, unmet needs, and workflow inefficiencies clinicians experience when using existing physiological monitoring technologies (e.g., bedside monitors, dashboards, and mobile systems).

- Insights from these interviews will inform the design of a mobile modular interface that better supports real-time decision-making, data interpretation, and clinical efficiency.

2. Research Questions

1. How do clinicians currently interact with physiological monitoring systems in hospital settings?
2. What are the main challenges or frustrations they face when using current tools (desktop, bedside, or mobile)?
3. How do they interpret, prioritize, and act on patient data from these systems?
4. When and why do clinicians switch between devices or interfaces (e.g., desktop → mobile → EHR)?
5. What features or data visualizations would improve their experience with mobile access?

3. Methodology

- Approach: Qualitative — Semi-Structured Interviews
- Duration: ~30–45 minutes per participant
- Mode: In-person (preferred) or virtual (via Zoom/Teams)
- Each interview will follow a flexible discussion guide to allow participants to expand on their experiences. Interviews will be audio-recorded (with consent) and transcribed for thematic analysis.

4. Participants

Group Role	Examples	Target #	Rationale
Clinicians	OB/GYNs, Nurses, Anesthesiologists, Residents	6–8	Direct users of patient monitoring systems.
Biomedical technicians	Clinical engineers, IT specialists	2–3	Understand system integration and data flow.
Hospital administrators (optional)	Nurse supervisors or department leads	1–2	Understand systemic workflow and adoption barriers.

Total: 8–12 participants

Recruitment criteria:

- Currently working in hospital or outpatient settings.
- Regularly use digital monitoring tools (e.g., ECG, pulse oximeter dashboards).
- At least 1 year of experience in clinical practice.

Recruitment process:

- Participants will be recruited through a combination of professional and academic networks.
- Connections will be established via a nursing professor contact, local IU Health partnerships, and personal professional networks.
- Recruitment messages will be shared through email or LinkedIn, outlining the study purpose, time commitment, and consent process.
- A snowball approach will be used, inviting participants to refer to qualified colleagues.
- All participants will provide informed consent before participation.

Recruitment Template:

Subject: Invitation to Participate: Interview on Improving Clinical Monitoring tools for ANS

Body:

Hello [Participant Name]

I'm reaching out as part of a Capstone Project at IUI exploring how clinicians and clinical support staff use physiological monitoring systems in hospital and outpatient settings. Our goal of this study is to identify pain points and unmet needs with current monitoring tools to help inform the design of a more intuitive mobile interface that supports real-time decision-making and efficiency.

We're inviting clinicians and clinical support professionals (e.g., nurses, OB/GYNs, anesthesiologists, biomedical technicians) to participate in a 30–45 minute semi-structured interview, conducted in-person or virtually at your convenience. Conversations will be recorded with your consent and kept completely confidential. Your insights will directly shape the next generation of clinical monitoring design.

If you’re interested or would like to have more details, please reach out to this email or contact me at [preferred email address].

Thankyou for your time and consideration,
[Your Name]

5. Interview Structure

For Clinicians:

Section	Question
A. Warm-up & Context Setting (5 min)	Can you briefly describe your current role and how long you’ve been practicing in this setting? What devices or systems, if any, do you use in your routine workflows to monitor patients’ vital signs or physiological status? Do you work with any high-risk populations such as patients with preeclampsia or gestational hypertension? If so, what data points become most important in those cases?
B. Current Workflow & Interpretation (10–15 min)	Can you walk me through how you typically check and interpret a patient’s physiological data? What types of tools or technologies (e.g., bedside monitors, desktop systems, mobile tools) do you use, and how do you decide which to use in different situations?

	How do you usually receive information about patient changes (e.g., alarms, notifications, chart updates)? Are there any metrics—standard or advanced—that you find particularly useful in understanding a patient’s physiological status? If you were presented with more complex data (such as variability metrics), how would you prefer that information to be displayed to support quick interpretation, if at all? When reviewing continuous data, do you find trends, single values, or another format most helpful? Are there indicators you pay attention to when watching for early signs of physiological distress? If so, what do those look like in your workflow?
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**C. Pain Points,
Challenges & Gaps
(10–15 min)**

How would you describe your overall experience with the monitoring tools you use today? Are there aspects that work especially well or aspects that are challenging?

Has technology ever affected your ability to respond to patient changes? If so, can you describe an example?

What has your experience been with alerts and alarms in your current workflow?

If a new alerting system were introduced for early physiological changes, what would help it feel useful and not overwhelming?

How does the distribution of patient data across different devices or interfaces affect your work, if at all?

Are there any barriers that make continuous monitoring difficult during times such as shift transitions or postpartum discharge? If so, can you describe them?

D. Technology, Mobility & Patient Communication (10–15 min)	In what situations, if any, do you find mobile access to patient data helpful? If you use a smaller screen (phone, tablet, etc.), how should data be displayed to promote rapid support and accurate interpretation? When you communicate trends or concerns to patients or families, how do you typically explain physiological changes? Are there aspects of that communication that you find challenging? What would your thoughts be about a system that uses AI or predictive analytics to identify early signs of deterioration? What information would help you feel comfortable acting on such alerts?
E. Wrap-up (5 min)	If you could change or improve one aspect of your current monitoring workflow or tools, what would you focus on improving? What would a continuous, real-time physiological monitoring platform need to offer for it to be helpful in your work? When integrating new technologies/tools in your workflow, who are the people that typically make the decision to approve it?

For women who have experienced or undergoing pregnancy:

Section	Question
A. Warm-up & Background (5 min)	How far along are you in your pregnancy, and how has your experience been so far? What sources do you usually rely on for information about your health during pregnancy (e.g., OB/GYN, apps, online resources, family)? When you think about monitoring your health during pregnancy, what matters most to you?
B. Understanding of Symptoms & Body Awareness (10–12 min)	In what ways, if any, do you keep track of changes in your body—such as blood pressure, heart rate, swelling, dizziness, or general discomfort? Are there any symptoms or changes that you tend to pay more attention to? If so, what makes those stand out? What information have you received about signs to watch for during pregnancy, if any? How do you feel about your ability to recognize those signs? Have there been times when you wanted more information or reassurance about something you were experiencing? If so, can you describe those moments?

C. Current Tools & Monitoring Experience (8–10 min)	Do you use any tools—such as apps, wearables, or medical devices—to monitor your health? If so, what has your experience been with them? How do you usually access your health information, such as readings, lab results, or appointment notes? How well do the tools or resources you use meet your needs? Are there any gaps or things you wish were different?
D. Anxiety, Safety, and Communication Needs (10–12 min)	Have there been any situations during your pregnancy that felt challenging or required additional support? If so, can you tell me about them? When you notice a new symptom or change, how do you usually decide what to do next? How would you describe the way your healthcare providers communicate information about what’s happening in your body and why? If you were to receive explanations or educational information through technology, what formats would work best for you (e.g., messages, visual charts, summaries)?

E. Technology Expectations & Feedback (10–12 min)	If a mobile app or wearable monitored your health, what kinds of information would be most useful for you to see? If the app needed to send you alerts or notifications, what would be the best way for you to receive them? Would you prefer seeing detailed information, summaries, or a combination? How do you feel about an app using AI to notify you if something looks unusual? What factors would influence your level of trust in it?
F. Wrap-up (3–5 min)	If you could change or enhance anything about your pregnancy monitoring experience, what would it be, if anything? Is there anything else you wish you had during this pregnancy to help you feel safer, more informed, or more supported?

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6. Data Collection & Analysis

- Recording: Audio or video (with consent)
- Transcription: Automated, verified manually
- Analysis Method: Thematic analysis using codes such as workflow barriers, cognitive load, alert management, data visualization, trust in automation, and mobile usability.
- Tools: Figma, Microsoft Excel, or Miro for affinity mapping.

Deliverables:

- Thematic summary report of recurring pain points
- Journey map of clinician-system interactions
- Insight statements (e.g., “Clinicians need...”)
- Early design implications for modular mobile UI

7. Ethical Considerations

- Obtain informed consent from all participants.
- Anonymize identifiable data in transcripts and reports.
- Securely store recordings and transcripts.
- Ensure participants understand that their insights will inform interface design, not clinical decisions.

8. Timeline (Estimated 3 Weeks)

Week	Activities
1	Prepare interview guide, recruit participants, secure IRB or departmental approval
2	Conduct 8–12 interviews (in-person or online)
3	Transcribe, code, and synthesize findings into key insights and design opportunities

Cognitive Walkthroughs

A cognitive walkthrough is a technique used to evaluate the learnability of a system from the perspective of a new user. We prepared a series of questions to walk the researchers through the sensor fitting and onboarding process to identify aspects of the flow that could be challenging to new users.

The walkthrough was conducted by research specialists with extensive experience in managing physiological data and laboratory equipment:

- **Clinical Research Coordinator:** Responsible for subject recruitment, IRB protocol documentation, and managing the data pipeline between researchers and institutional review boards.
- **Laboratory Research Specialist:** Responsible for managing hardware, developing data processing scripts, and training students on maintaining high-quality physiological signals

Cognitive Walkthrough Execution

Phase 1: Sensor Fitting & Orientation

- **Will users try to achieve the right result?**
 - **Yes:** Users want to be "ready" for the study, but they often struggle with the physical orientation of the strap.
- **Troubles & Behaviors:** A common failure occurs when users put the strap on **upside down**. While the sensor still reads, the inverted signal complicates peak-detection algorithms, often rendering the data confusing or unusable if not caught immediately.

Phase 2: Monitoring & Signal Maintenance

- **Will users know that progress is being made?**
 - **Yes:** Researchers use live signal feeds to confirm a "READY" state, but participants are often unaware if their behavior is degrading the data.
- **Troubles & Behaviors:** Users often display "**movement artifacts**"; even a simple sneeze or shift in posture can cause a loose strap to fall or lose contact. If the system doesn't provide immediate feedback, the user continues the protocol unaware that the "source of truth" has been compromised.

Critical Design Implications

Based on the identified behaviors and fitting troubles, the following requirements are essential for the TruePulse system:

- **Posture and Fitting Guidance:** The system should provide the user with adequate support to understand how to put the sensor on via visuals and communicate its constraints and features such as the absence of physical buttons.
- **Signal Quality Awareness:** Since users cannot "see" their own heart rate quality, the app must provide a clear "DEGRADED" or "READY" status to prompt them to sit still or tighten the strap.
- **Automated Landmark Logging:** To account for unavoidable behaviors (like sneezing or coughing), the tool should allow researchers to log these moments digitally so that analysts can differentiate between a biological event and a machine artifact later.

Analysis

Visual Analyses

Personas & User Journey Maps



Sarah Jenkins
Acute Care Clinician

About

- 38
- 12 hour shifts
- 10+ Yrs Exp.
- High-Noise Unit

Skeptical Pragmatic Patient-Focused

Time-Poor Empathetic Tech-Cautious

Motivations

Sarah is driven by patient safety above all else. She views technology purely as a tool to serve the clinical goal, not an end in itself. Her primary motivation is to efficiently filter out environmental and data "noise" so she can focus her limited time on true clinical crises and human connection at the bedside.

Goals

- Assess immediate patient status (Stable vs. Unstable) in under 5 seconds upon entering a room.
- Reduce time spent deciphering "black and blue" EMR data to increase time for patient interaction.
- Explain complex physiological data to anxious families in plain language without causing panic.

Pain Points

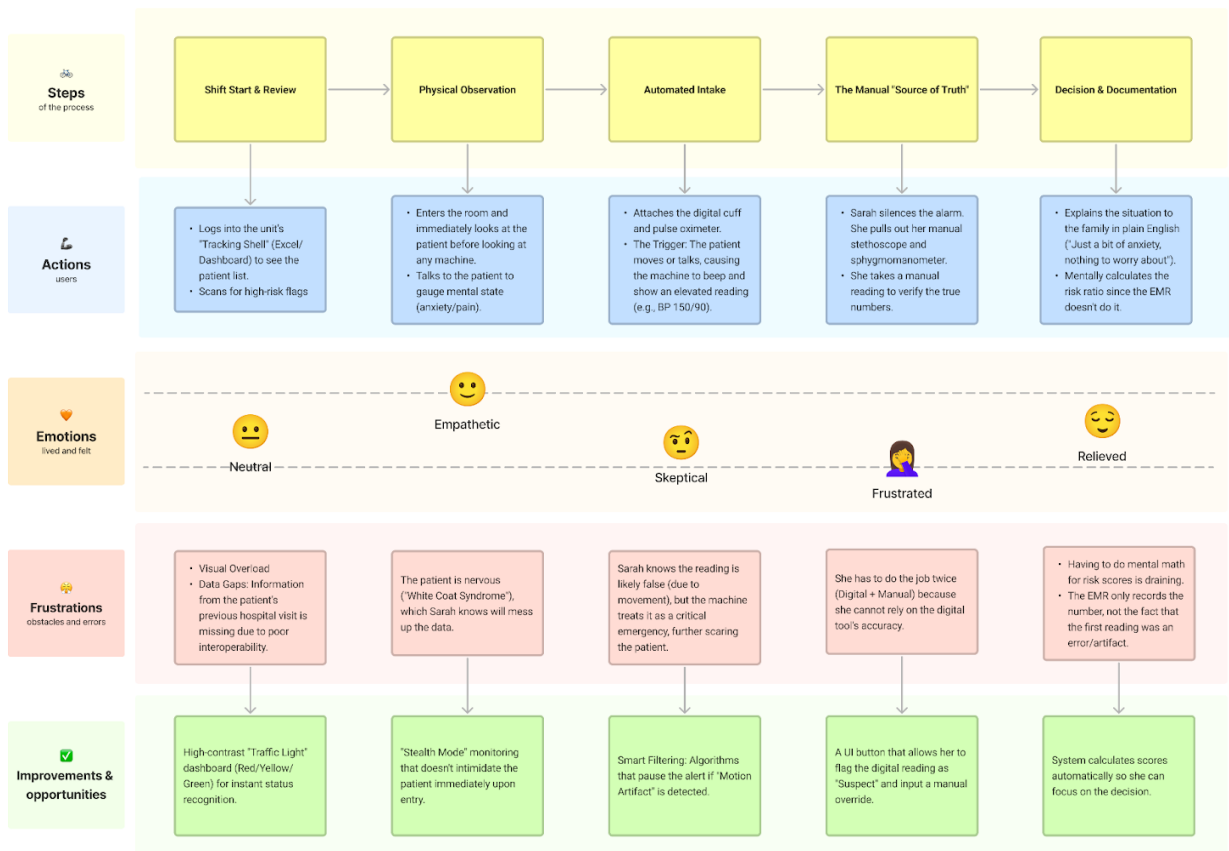
- **Alarm Fatigue:** She has become desensitized to constant monitor beeping because most alerts are caused by patient movement rather than real danger.
- **The "Black Box":** EMRs provide automated risk scores without showing the underlying math, forcing her to trust data she cannot verify.
- **Cognitive Overload:** Having to perform mental math for critical metrics (like Preeclampsia ratios) while under pressure at the bedside.

Tools & Technologies

- **Manual Stethoscope/Cuff:** Her absolute "source of truth" for validated data when digital tools seem off.
- **AI Predictions (Muse):** Used only as a loose guide for prioritizing patient lists, but rarely trusted blindly without human proofreading.
- **Bedside Monitors:** Used for general trend awareness (up or down arrows), but specific snapshot numbers are often ignored due to noise artifacts.

Behaviors & Workarounds

- **The Validation Loop:** If a digital cuff gives a critical reading, she immediately discards the data point and re-takes it manually to ensure accuracy.
- **Ignoring Protocol:** She delays logging data for anxious patients ("White Coat Syndrome"), waiting 10 minutes for them to calm down to get a "true" baseline reading.
- **Physical Corroboration:** She prioritizes looking at the patient physically (e.g., skin color, breathing ease) over monitor data if the two conflict.





Elena Rossi

Expectant Mother (High Risk)

About



32



28 weeks pregnant

Anxious

Proactive

Data-Sensitive

Community-Reliant

Condition:

- Gestational Diabetes & Pre-eclampsia Watch

Motivations

Elena is driven by the health of her baby above all else. She wants to feel "empowered" and in control of her pregnancy journey but is easily overwhelmed by raw medical data. Her primary motivation is to translate abstract numbers (BP, blood sugar) into actionable peace of mind, seeking human reassurance over algorithmic predictions.

Goals

- Establish Baselines:** Quickly understand "what is normal" for her specific body versus what is a generic pregnancy statistic.
- Actionable Synthesis:** Receive immediate, practical advice (e.g., "drink water," "rest now") rather than just seeing a red warning number or a graph she can't interpret.
- Bridge the Gap:** Efficiently communicate daily symptoms to her provider without waiting for the next monthly appointment or feeling like a burden.

Pain Points

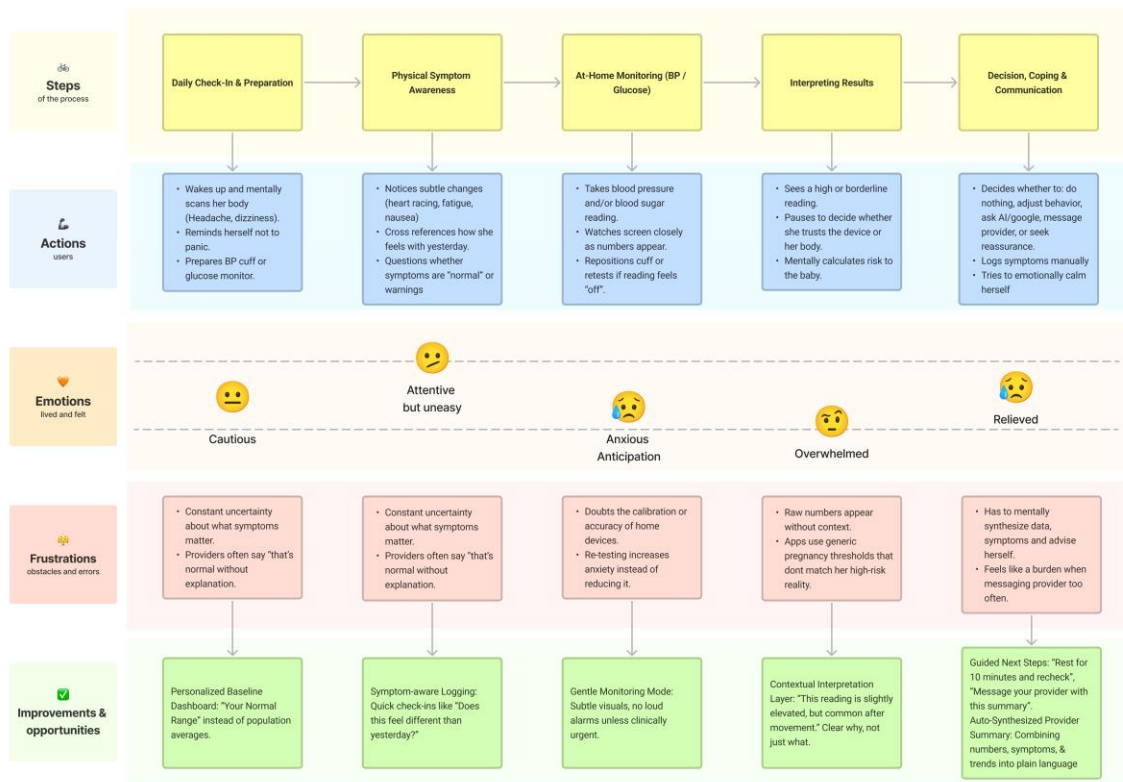
- Data Anxiety:** Viewing raw results (like blood sugar spikes or BP fluctuations) before a doctor explains them causes immediate stress and fear for the baby's safety.
- The "Normalcy" Void:** Medical providers often dismiss symptoms as "normal" without explaining the physiology, forcing her to self-educate through Google or family.
- Algorithmic Rigidity:** Pregnancy apps often fail because they rely on standard calculations (like standard LMP) that don't match her unique medical reality, leading her to abandon them.
- Alert Fatigue:** She becomes stressed or annoyed by loud, intrusive auditory alarms for non-emergencies; she prefers subtle nudges.

Tools & Technologies

- Home Monitoring Devices:** She relies on a BP cuff and Glucose monitor as her daily "source of truth," though she sometimes doubts their calibration or accuracy.
- AI as "First Responder":** She uses tools like ChatGPT for immediate triage of low-stakes questions ("Is this symptom normal?") before escalating to a human.
- Community Forums:** Used to validate her physical sensations against the shared experiences of other mothers, which she trusts more than generic app articles.

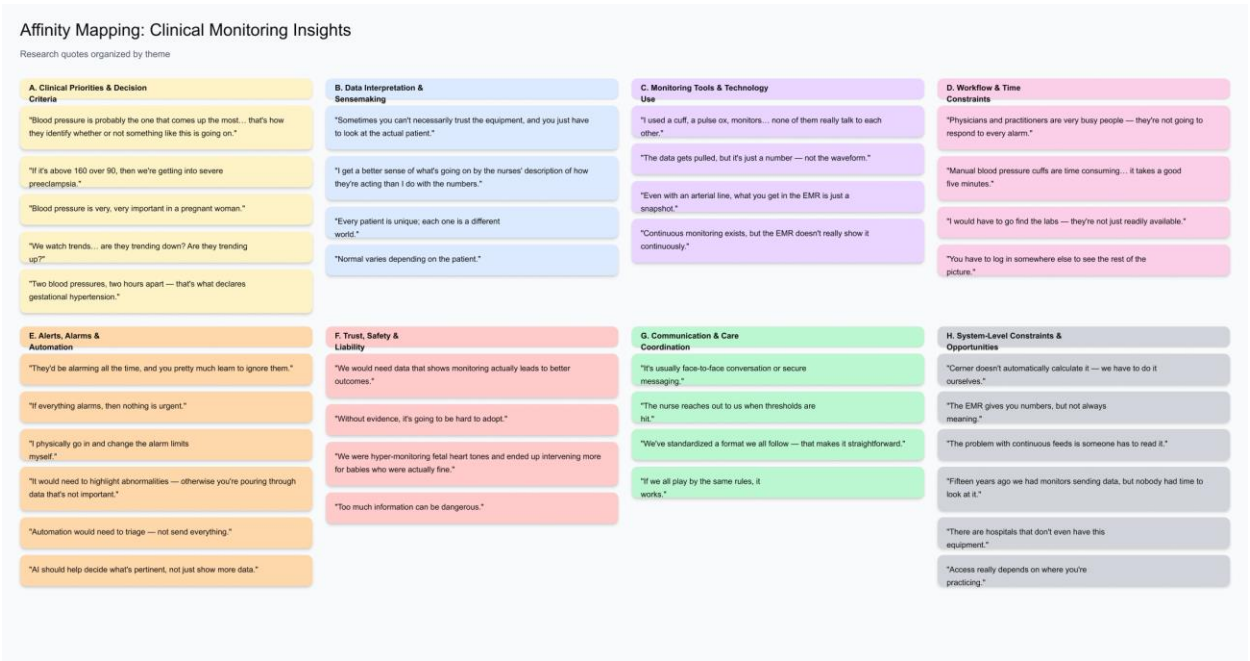
Behaviors & Workarounds

- The Validation Loop:** If a digital app gives a prediction that conflicts with how she feels physically, she prioritizes her somatic awareness (body sensation) and ignores the data.
- Hybrid Information Seeking:** She cross-references everything: first asking AI/Google, then verifying with a family member (often a medical proxy), and finally confirming with her doctor.
- Manual Synthesis:** She keeps physical notes or mental logs of symptoms to present to her doctor because she feels digital apps don't capture the full context of her condition.

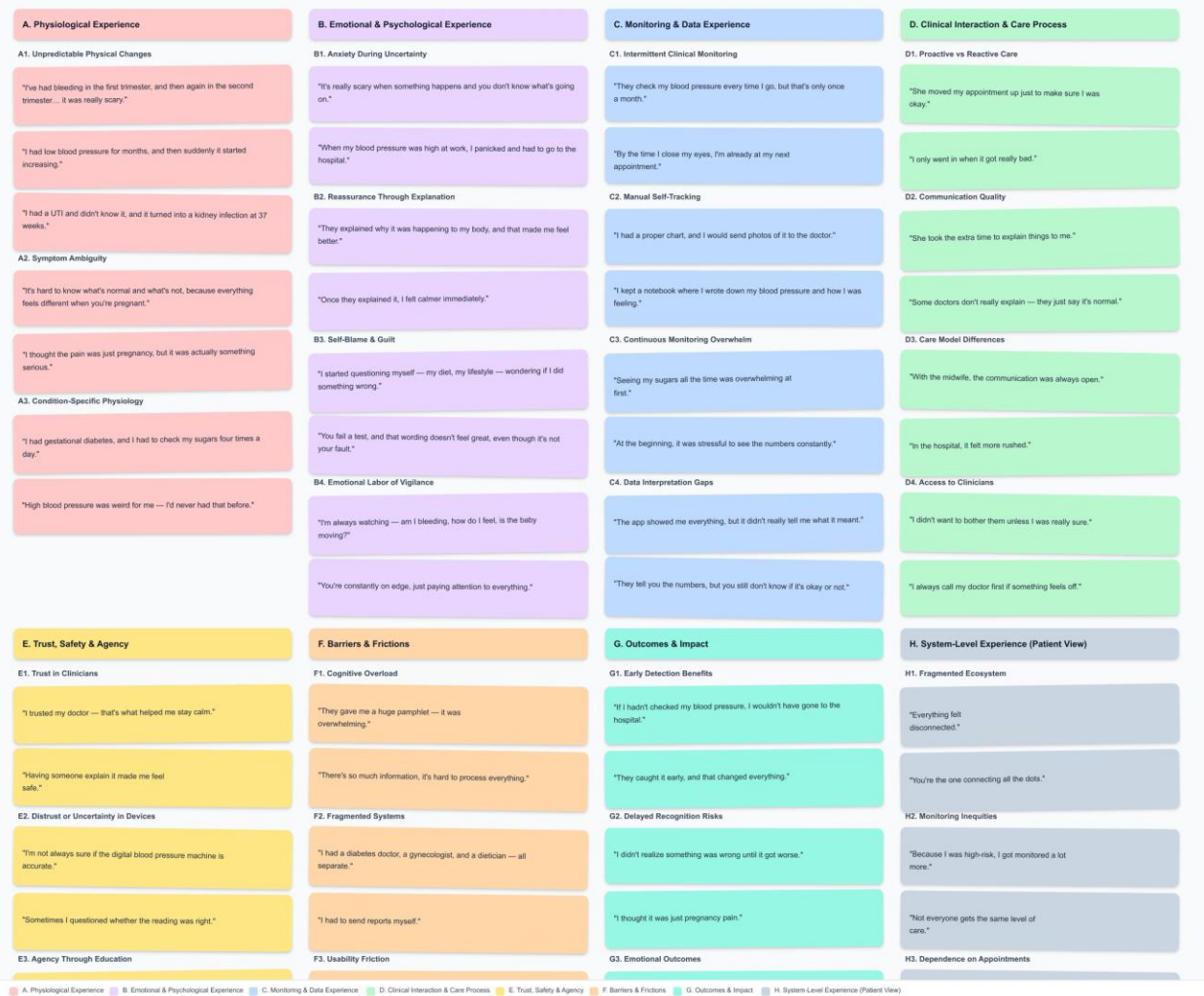


Affinity Mapping

Clinician Affinity Map



Pregnant Women Affinity Map



Link to Artifacts:

<https://docs.google.com/document/d/1VhGGUknfurWSTdwZzrb6E9ST3kWbn4fLV0NDKQjW2ns/edit?usp=sharing>

Outcome of Analysis: Functional Requirements

Application-level requirements

The application-level requirements describe the constraints and decisions about the entire application, not the individual user experience. The requirements are specified as follows:

- **Interface Metaphors**
 - “Clinical Dashboard” metaphor: A modular panel-based layout like hospital monitoring stations
 - “Vital Cards” metaphor: Each ANS metric (ECG, HRV, Respiration, BPV) represented as an interpretable “Card” like patient charts

- “Traffic Light Health States” metaphor: For quick status interpretation (green = stable/good, yellow = caution, red = unstable/bad)
- Interaction Types
 - Direct manipulation (tap, swipe, expand graph modules”
 - Monitoring Visuals (glanceable summary, monitoring graphs)
- Input Devices
 - Physiological sensors streams (ECG, respiration, pulse oximetry)
 - Touch input (tap, swipe, navigation inputs)
- Output Devices
 - Real time ANS visualizations (graph, and data summary)
 - Alerts and trend summaries
 - Sensor setup guide and error feedback
- Constraints
 - Must integrate with off the shelf sensors
 - Must remain hardware agnostic for future sensor additions
 - Must ensure HIPAA compliant data security
 - UI must remain neutral, clinical, and modular
- Budget
 - Moderate/low Budget: Primarily student led design & prototype
- Timeline
 - Research & Conceptualization → Nov-Dec
 - Low Fidelity & Testing → January
 - Mid Fidelity & Testing → February
 - High Fidelity & Testing → March
 - Final prototype & Handoff → April-May
- Developer Guidance / Design Principles
 - Standards: Use medical standard color semantics, clear contrast ratios, and non-ambiguous icons.
 - Style Guide:
 - Minimalist, information dense yet glanceable layouts
 - Expandable charts with progressive disclosure
 - Plain language explanations beside technical metrics
 - Human Factors:
 - Low cognitive load, especially for clinicians switching between tasks.
 - Use haptic and auditory alerts for rapid interpretation
 - Avoid overwhelming users with simultaneous data streams, prioritizing critical signals
- Context of Use
 - Indoor clinical environments: Hospitals, Outpatient labs, OB/GYN clinics
- User Tasks / User Goals to be Supported

- Clinicians Need to:
 - View real time ANS data
 - Receive prioritized alerts for abnormal patterns
 - Validate if data is accurate or artifact heavy
 - Add contextual notes or mark clinical events
 - Understand patient trends over time
- Biomedical Technicians Need to:
 - Check sensor connectivity
 - Troubleshoot sensor errors
 - Verify calibration
- Administrators Need to:
 - View multiple patients at once
 - Track alerts across a ward
- Use Cases
 - Use Case 1 - Sensor Setup & Validation
A clinician attaches ECG and Respiration sensors to a patient, and the app verifies correct placement with real time feedback
 - Use Case 2 - Real Time Monitoring
The clinician glances at the mobile dashboard to see trends in HRV, Respiration, and ECG.
 - Use Case 3 - Alert Prioritization
The app identifies abnormal ANS patterns (sympathetic overactivation) and issues a ranked alert.
 - Use Case 4 - Artifact Detection
The system flags noisy or unreliable signals and offers guidance
 - Use Case 5 - Trend Interpretation
In conjunction with AI it provides a plain language summary to analyze trends.

Functional Requirements

Functional requirements define what the system must enable for both pregnant women and clinicians to support effective interaction with TruePulse's ANS-tracking technology. These requirements describe the essential capabilities the interface must provide to ensure clarity, usability, and meaningful decision-making.

- **As a pregnant woman without medical training**, I want the interface to **translate ANS/HRV data into simple categories** (e.g., "Stable," "Elevated Stress," "Needs Attention") so that I can **understand what is happening** in my body without confusion.
- As a pregnant woman, I want to see how my behaviors (sleep, activity, hydration, stress episodes) relate to changes in my ANS data so that I understand what might have triggered a shift.

- As a pregnant woman who may feel dizzy or fatigued, I want to record a symptom episode quickly with minimal interaction so that I can capture what I am experiencing without struggling through a long form.
- As a pregnant woman, I want the app to clearly explain how my data is being used and what each metric means so that I can trust the recommendations being provided to me.
- As a pregnant woman experiencing stress, I want the system to recommend simple, clear actions (such as breathing exercises or hydration reminders) so that I know what steps to take to support my wellbeing.
- As a clinician caring for pregnant women, I want to receive a concise weekly summary of key ANS trends and flagged events so that I can quickly determine whether follow-up or intervention is needed.
- As both a pregnant woman and a clinician, I want symptom logs to align with physiological data trends so that both perspectives support clear communication and shared decision-making.

Describe Opportunity and Recommendations

Design opportunities highlight the gaps, needs, and areas where the system can create meaningful improvements or new value for users.

- There is a clear opportunity to build an interface that translates complex ANS and physiological data into simplified, meaningful insights for pregnant women, reducing cognitive load and improving self-monitoring during pregnancy.
- The project can support stronger communication between pregnant women and clinicians by aligning subjective symptom logs with objective physiological trends, enabling more informed, shared decision-making.
- Providing contextual explanations and transparency around data interpretation offers an opportunity to increase trust and long-term engagement with the device.
- A unified timeline that combines behavior, symptoms, and ANS patterns allows the system to identify personalized triggers, creating opportunities for more targeted, proactive interventions.
- For clinicians, there is an opportunity to streamline prenatal care with concise summaries and optional deeper views of physiological detail, reducing time burden while enhancing clinical insight.

Design recommendations outline the specific design approaches or solutions the system should adopt to address user needs and improve the interface experience.

- Design the user facing interface to prioritize clarity, using color coding, natural-language summaries, and simple categorizations ("Stable," "Elevated Stress," etc.) to make ANS patterns understandable without medical background.

- Implement a rapid symptom-logging mechanism such as single-tap tags and large touch targets to ensure pregnant women experiencing discomfort can record episodes quickly and accurately.
- Incorporate behavioral overlays (sleep, hydration, stress moments) directly onto the ANS trend timeline to help users identify relationships between daily habits and physiological changes.
- Add trust-building elements such as “Why am I seeing this?” explanations, definitions of metrics, and transparent descriptions of how recommendations are generated.
- Provide actionable, supportive recommendations (e.g., breathing exercises, hydration reminders) that are tailored to the user’s current physiological state.
- For clinicians, design a dashboard that begins with a high-level summary (key risks, number of alerts, major trends) with the option to expand into deeper data only when needed supporting efficient review without overwhelming detail utilizing a progressive disclosure pattern.
- Create a shared view of symptom logs and physiological patterns so clinicians and pregnant women can have aligned discussions during appointments, reinforcing collaboration and trust in the system.

Key Research Findings

Ideation

Information Architecture

The interface is organized around a protocol-first workflow. Where the users prepare sensors, complete a guided protocol, then receive and review results. The information architecture prioritizes clarity, data quality awareness, and low cognitive load, especially during active recording.

Architectural Rationale

This IA encompasses our user research and aligns it with the company's vision for a data collection app for Mothers.

- Data Integrity Gate: Every session is gated by a real-time signal quality check (READY/DEGRADED/ERROR).
- High-Fidelity Context: Mood surveys (PHQ-9/WHO-5) are timestamped with physiological data to provide clinicians with biopsychosocial context.
- Hands-Free Interaction: Since the phone sits on a table, the UI relies on high-visibility motion and audio chimes to guide the user through the 3-1-1-1-3 sequence.

Task Analysis (Flow) - Pregnant Women

Mothers

The app's navigation is structured around the user journey, ensuring a focus on trust, low cognitive load, and actionable, synthesized data. I.E. Entry Point & Onboarding Flow This flow is designed to quickly establish user identity, consent, and necessary medical context, addressing concerns around data privacy and algorithm inflexibility.

IA Area / Screen	Displayed Data	User Input (Activity)
SSO Login (Entry Point)	Hospital/Partner Branding, App Name, "Sign in with [SSO Provider]" Button, Privacy Policy Link, Terms of Service Link.	Tap on "Sign in with [SSO Provider]" (e.g., Google, Apple, or Institutional SSO).
Consent Management	Mandatory EMR/Privacy Policy text explaining what data is collected, how it's used/stored, and compliance (e.g., HIPAA).	Mandatory tap on "Accept & Continue" (or "Decline," which exits the app).
Account Creation	Input fields for basic demographic data.	User enters Name, Age, Gender, Region. User selects Reason for using the app (e.g., Pregnancy, Postpartum).

Clinical Linkage	Input field or search functionality for medical providers.	User inputs or selects Consulting Doctor/Hospital name/code.
Device Connection	Guided, illustrated setup instructions for linking tracking devices.	User taps a button to Connect Apple Watch or other commercial sensor combo. User follows pairing steps.
Pregnancy Data Input	Input fields for medical history and current pregnancy status (Pulls from EMR if linked).	User inputs or confirms: Pregnancy Phase, Expected Delivery Date, Last Check-up Details, Last Reported Baby Vitals, Next Appointment Date, Doctor's Insights.
Partner Access	Input field to invite a partner/caregiver.	User enters Partner's email/ID to send an invitation; User taps to Manage/Revoke access.

II. Core App Navigation

Once onboarded, the user is directed to the Home Screen. The main navigation branches into the core functions: **Home**, **History/Tracking**, and **AI Support**. 1. Home Screen (Human-Centered Dashboard)

The screen prioritizes synthesized status and actionable suggestions over raw metrics (Addresses P3/Elena Rossi: Anxiety from uninterpreted raw data).

IA Area / Component	Displayed Data	User Input (Activity)
Current Status	Single, clear status indicator (e.g., Green/Blue for Calm, Yellow for Attention).	Tap on the Status Indicator to view a quick, synthesized explanation of the current status.
Status Assessment & Action	Contextual suggestion based on current vitals: "Doing good, " "Needs to rest, " "Needs attention from doctor, " or "Quick fixes" (e.g., "Time to hydrate").	Tap on the Suggestion to dismiss or view a related educational link.
Immediate Logging	Quick-access buttons/widgets.	Tap "Log Kicks", "Start Contractions Timer", or "Log Mood/Symptoms."
Current Vitals	Mother's current vitals (e.g., Heart Rate, BP) displayed in a small, non-dominant area.	None (Display only).

Apple Watch Widget	Simplified status (e.g., "Doing Good") and a quick prompt (e.g., "Log Kick") on the wrist	Physical Tap on the watch screen/complication.
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2. Session History & Tracking

This area is designed for user reflection and consolidation of all Mother and Baby progress.

IA Area / Screen	Displayed Data	User Input (Activity)
Session List (Landing)	Chronological list of past tracking sessions and logged events (Date, Protocol Name, High-Level Result).	Tap on any Session to view detailed results.
Tabbed Navigation	Tab selector: "Mother's History" / "Baby's Journey."	Tap to switch between Mother and Baby tabs.
Mother's Historical Tracking	Vitals and tracking items (HRV, RHR, BP) over weeks/months (as charts/graphs). Context-aware educational overlays (e.g., "what is developing this week").	User scrolls/zooms on charts; Taps on educational overlays to expand information.
Baby's Journey Tab	Fetal Development: Baby Formation Stage with Visual Representation (based on week of pregnancy). Clinical Data: Last Reported Baby Vitals, Expected Delivery Date. Activity Log summary view.	Tap on an activity (Kicks, Contractions, etc.) to go to the Activity Log Detail Screen.
Activity Log Detail Screen	Historical log and visualizations for a specific activity (e.g., Kick Count trend, Contraction duration/frequency, Belly Size log).	Tap a button to Log New Entry (for Kicks, Contractions, Early Lactation, or input inches for Belly Size).

3. AI Support & Information Triage

A conversational interface that acts as a first-line triage tool before escalating to a human expert (Axis 2: Hybrid Intelligence).

IA Area / Screen	Displayed Data	User Input (Activity)
AI Q&A	Conversational chat window with previous AI interactions	User types detailed or vague questions into the

	and suggested prompts.	chat field.
AI Guidance & Suggestion	AI-generated answers, followed by a calm suggestion on next steps.	User can provide feedback on the answer (e.g., "Helpful/Not Helpful").
Community Link	A prominent link to a community forum.	Tap to Navigate to the Community Forum.

1 Screen 1: Login & Research Onboarding

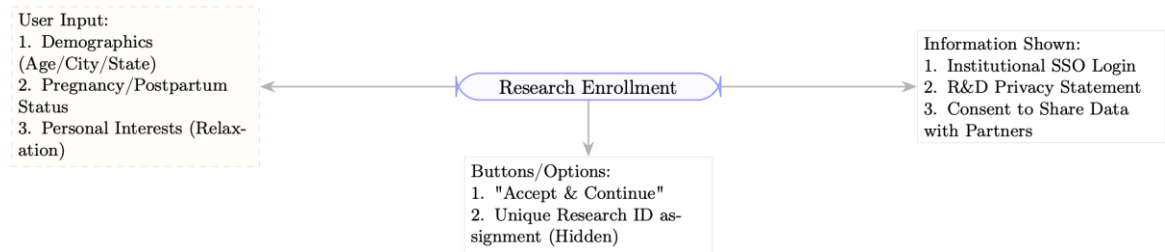


Figure 1: Onboarding: Establishing the Research Identity

2 Screen 2: Sensor Readiness Hub

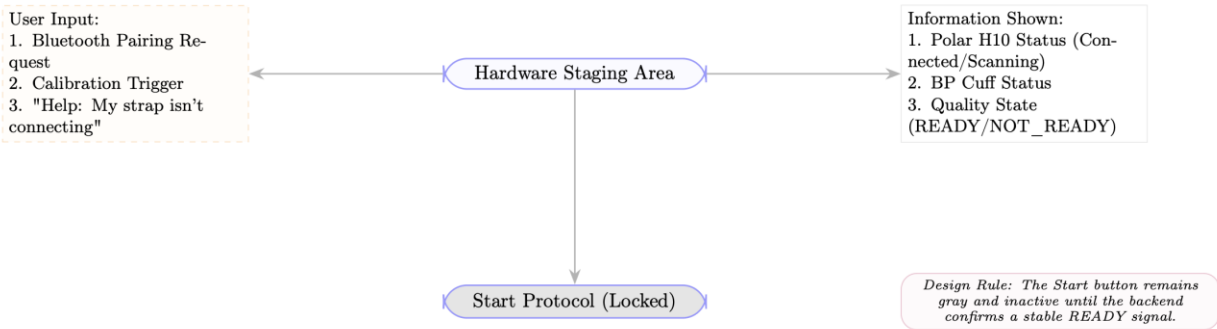


Figure 2: The Data Gate: Enforcing Technical Compliance

3 Screen 3: Home Dashboard & Knowledge Hub



Figure 3: Daily Operation: Balancing Information and Action

4 Screen 4: Biopsychosocial Survey

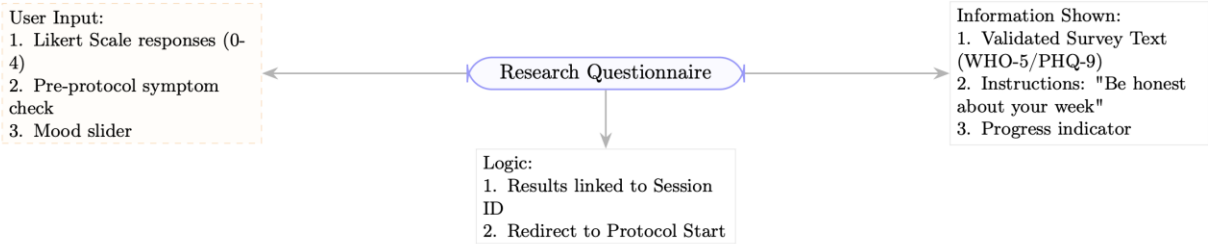


Figure 4: Contextual Data: Linking Mental State to Vitals

5 Screen 5: 3-1-1-1-3 Active Protocol

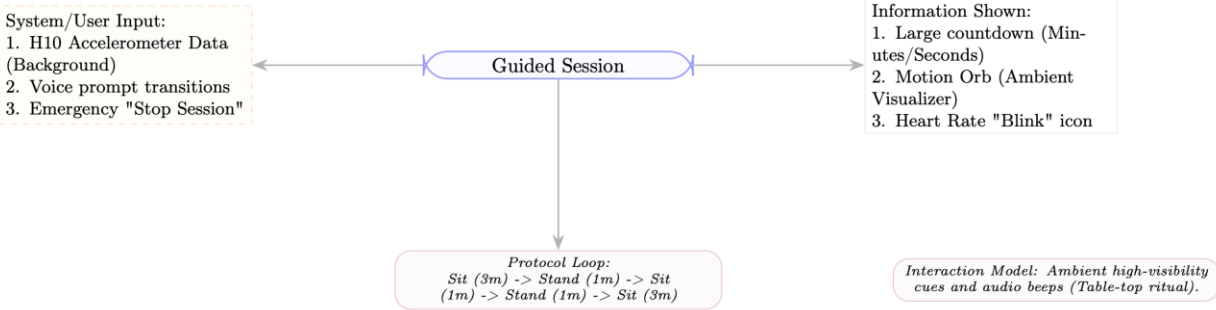


Figure 5: Execution: The Core 3-1-1-1-3 Sequential Data Collection

6 Screen 6: Session Verification & Review

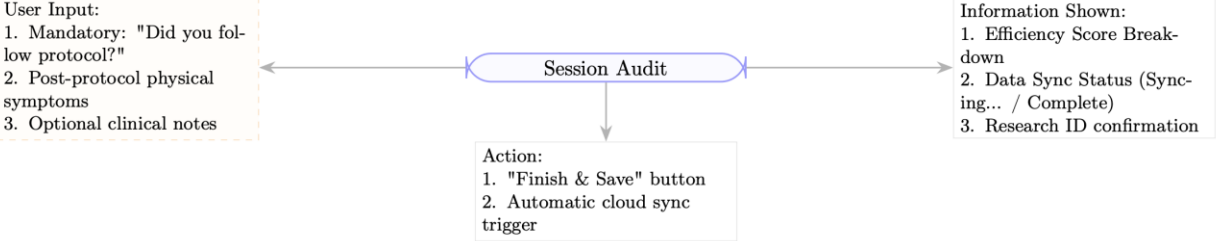


Figure 6: Closing the Loop: Data Submission and Quality Audit

7 Historical Data & Session History



Figure 7: Longitudinal Tracking: Managing the Research Record

Initial Sketches/Low Fidelity

The low-fidelity design phase focused on translating the project’s Information Architecture and user research insights into an early visual structure for the mobile interface. These sketches prioritized

workflow clarity, sensor readiness validation, and minimal cognitive load for users performing the ANS monitoring protocol. Because the application will be used while the participant is physically connected to physiological sensors, the design needs to support calm interaction, clear instructions, and immediate feedback on sensor integrity.

The early sketches explored the complete protocol journey, from onboarding to session completion. These screens were intentionally kept simple and monochromatic to allow the team to evaluate layout, hierarchy, and navigation before introducing visual styling. The primary goal was to ensure that users could progress through the protocol with minimal confusion while maintaining awareness of sensor status and data quality.

The low-fidelity prototype included the following core flows:

1. Onboarding and Consent

The onboarding flow introduces the study context and collects minimal participant information required for the research protocol. Screens such as the Privacy Policy and Account Creation establish trust and transparency by clearly explaining how physiological data will be collected and used. This step also gathers basic demographic details and contextual information such as pregnancy or postpartum status and user interests.

2. Sensor Connection and Readiness Validation

Because accurate physiological data is essential for ANS monitoring, the onboarding process includes a sensor connection step where the participant pairs wearable devices such as the Polar H10 ECG chest strap. The interface confirms when sensors are successfully connected and provides simple troubleshooting guidance if signals are not detected.

3. Home Dashboard

The home screen acts as the primary entry point for initiating the ANS testing protocol. It provides a clear “Start Protocol” action and quick access to educational resources explaining why physiological monitoring is performed. The layout emphasizes a single primary action to reduce cognitive load and help participants confidently begin the test.

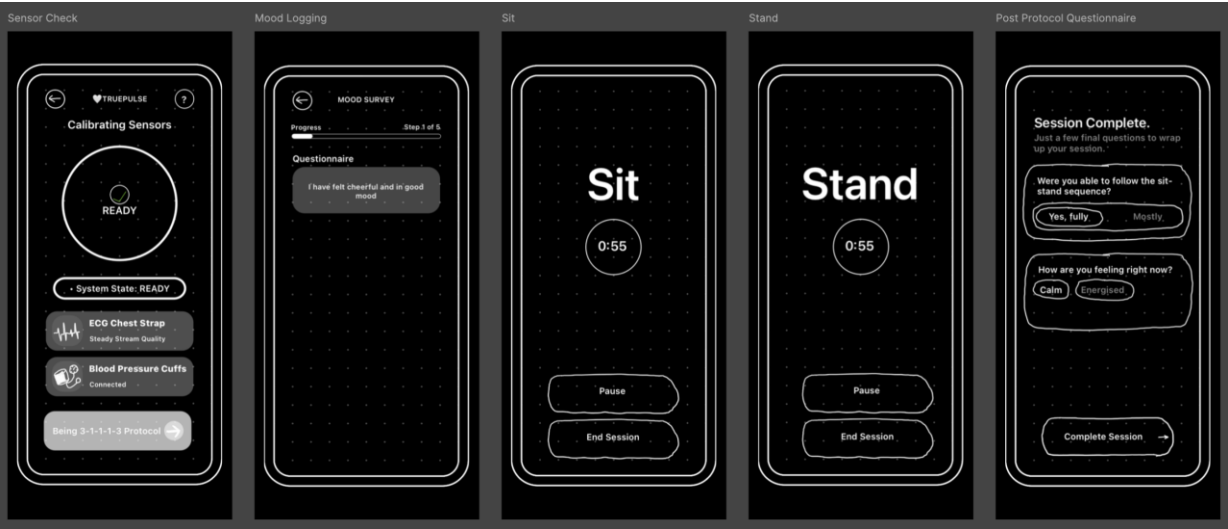
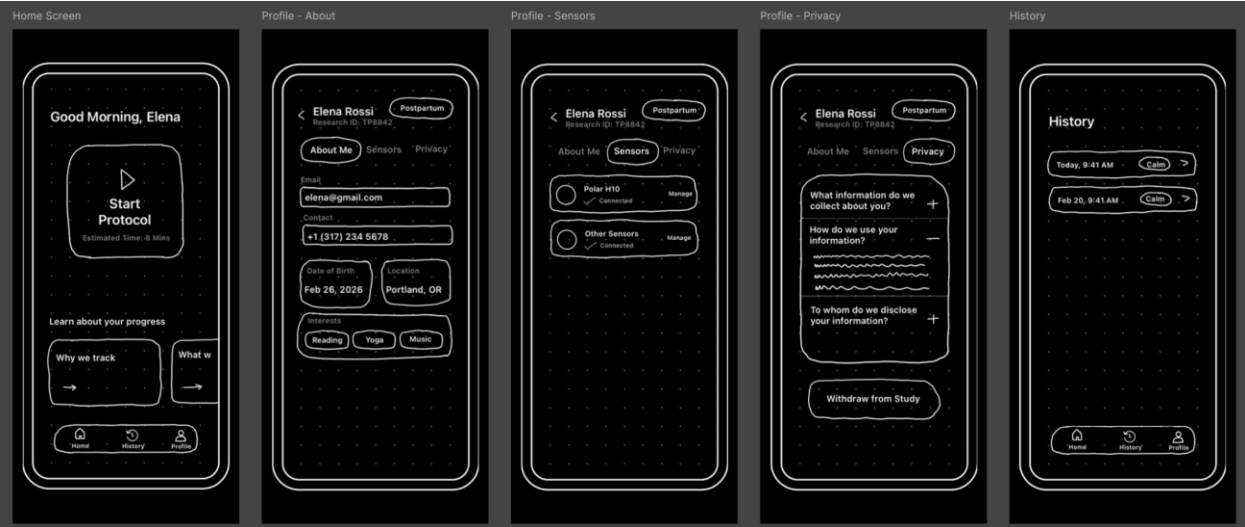
4. Profile and Sensor Management

Additional screens allow users to view profile details, manage connected sensors, and review privacy information. These sections were designed to support transparency and allow participants to verify which devices are actively streaming physiological data.

5. Session History

A history screen provides a simple chronological list of previous sessions. This allows participants and researchers to review previous recordings and maintain continuity between sessions.

Overall, the low-fidelity stage allowed the team to validate that the interface structure aligned with the protocol-first information architecture, ensuring that each step—from onboarding to session history—supports the goal of accurate physiological data collection with minimal user friction.



Mid Fidelity Prototypes

During the mid-fidelity stage, the team translated the earlier wireframes into a structured visual interface using Figma. This stage focused on refining the interaction flow, establishing a consistent design system, and validating how users would move through the application while maintaining clarity and emotional comfort during physiological monitoring.

At this point in the design process, all required screens for the core application workflow were implemented in Figma and prepared for interactive prototyping. The team also defined a foundational design system including color palettes, typography, spacing rules, icons, and reusable components, enabling consistent UI patterns across the application. The visual language was aligned with Google's Material Expressive 3 design principles, which emphasize soft color gradients, rounded shapes, and approachable interface elements that promote emotional comfort and trust for users interacting with health technology.

This design direction was intentionally chosen because participants may already feel anxious while wearing physiological sensors and participating in a health protocol. The use of calming gradients, soft curves, and simplified visual hierarchy helps reduce stress and maintain a sense of reassurance throughout the experience.

Onboarding and Sensor Setup

The onboarding flow guides the participant through the necessary steps to begin using the application and prepare the physiological monitoring devices. The process begins with a privacy policy screen where users can review information about data collection and consent to participate in the study.

Following consent, the application collects basic profile information such as the participant's name, phone number, pregnancy status (pregnancy or postpartum), and personal interests. This information provides contextual background for the monitoring session and allows the system to personalize the interface.

The next step in onboarding focuses on sensor setup, where users connect the Polar H10 ECG chest strap. The interface scans for available devices and confirms when a sensor has successfully paired with the application. Once the connection is verified, a setup completion screen displays a clear readiness indicator along with a brief summary of the user's profile and connected devices. This ensures that all required sensors are active before the participant begins a monitoring session.

Home Dashboard

After setup is completed, users are directed to the Home Dashboard, which serves as the primary navigation hub of the application. The dashboard emphasizes a single primary action: starting the sit-stand protocol. The central "Start Protocol" card provides a clear visual entry point into the physiological testing workflow.

Additional information cards below the primary action provide educational context, such as explaining why the system tracks posture transitions and how physiological signals relate to autonomic nervous

system activity. These elements help build transparency and user trust by explaining the purpose behind the monitoring process.

Navigation within the application is structured using a bottom navigation bar that allows users to move between the Home, History, and Profile sections.

Profile and Sensor Management

The Profile section allows users to view and manage their personal information, sensor connections, and privacy settings. The profile is organized into tabbed sections that include About Me, Sensors, and Privacy.

The “About Me” section displays participant details such as age, location, email address, and current maternal health focus. The “Sensors” tab lists connected physiological devices and allows users to manage their device connections if necessary. The “Privacy” tab provides clear access to the privacy policy and includes an option to withdraw from the study.

This structure ensures that participants always have transparent access to their personal information and sensor configuration.

Session History

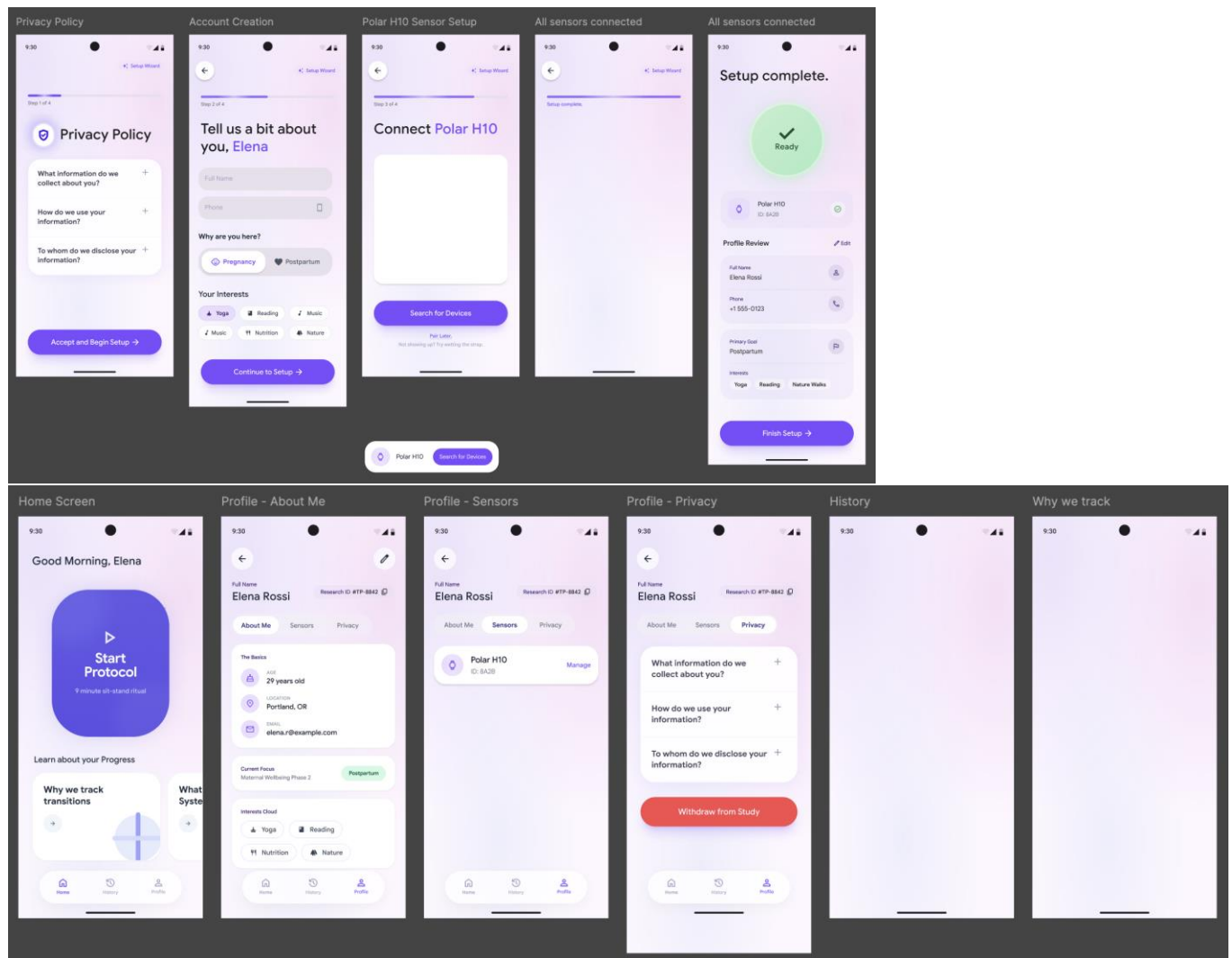
The History screen allows users to review previous protocol sessions and monitoring activity. While the mid-fidelity design currently presents a simplified layout, the intention of this screen is to support reflection and review of past sessions once physiological data visualization is implemented in later stages of the design.

Design System and Visual Language

The mid-fidelity prototype also introduced a structured design system to ensure visual consistency across screens. This system includes reusable UI components such as cards, buttons, navigation elements, and status indicators

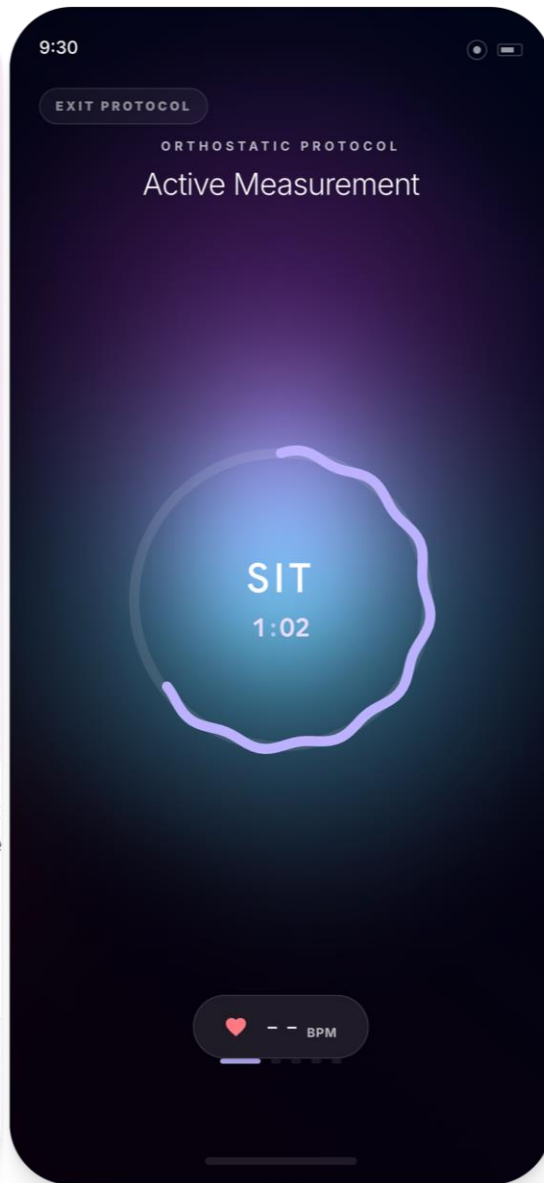
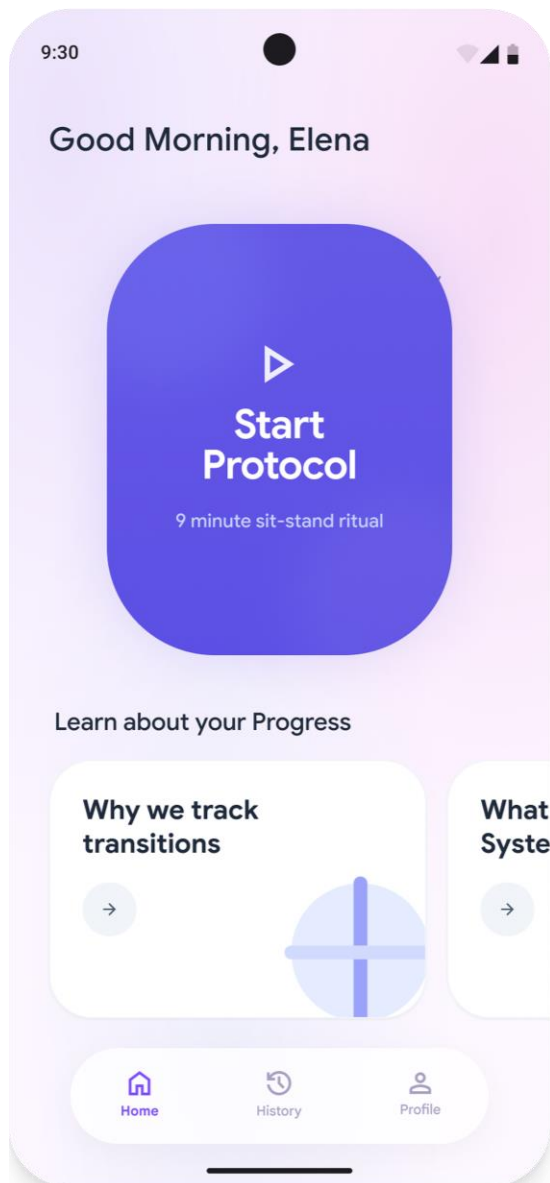
The color palette uses soft gradients and neutral backgrounds to maintain readability while conveying a calming visual tone. Rounded interface components and expressive color transitions are inspired by Material Expressive design principles, which aim to create interfaces that feel approachable and emotionally supportive. This is particularly important for health applications where users may experience anxiety related to their physiological state or monitoring devices.

By combining clear interaction flows with a calming visual design language, the mid-fidelity prototype demonstrates how the application can support both technical data collection and user emotional comfort during ANS monitoring sessions.



High Fidelity Prototypes

The high-fidelity screens were developed with the support of AI-assisted design tools, allowing us to rapidly iterate on visual, interaction, and data presentation components. This approach enabled us to create a realistic and fully functional prototype that could integrate with the Polar H10 sensor provided by our industry partner. By leveraging AI in the design process, we were able to simulate real-time physiological data, refine user flows, and ensure the interface supported accurate execution of the sit-stand protocol. This level of fidelity was critical in enabling meaningful usability testing, as participants could interact with the system in a way that closely mirrors the intended final product.





PROTOCOL COMPLETE

Recovery Insight

PEAK HR



75 BPM

↑ 22% vs baseline

HRV (RMSSD)



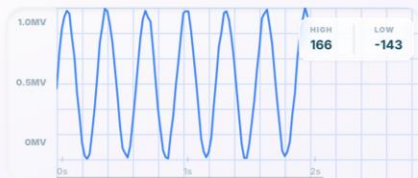
45 MS

≈ Standard variability



ECG Trace

512 Hz

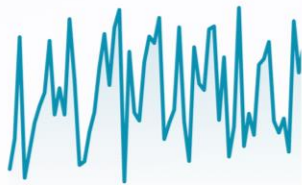


Heart Rate Response

3-1-1-1-3 MATERNAL PROTOCOL ANALYSIS

MAX
80

MIN
70



Usability Testing

Usability Testing Methods

Define and describe the methods

1. Cognitive Walkthrough Evaluation

We conducted an internal evaluation using a Cognitive Walkthrough to assess how easily first-time users can complete key tasks in the app. The team stepped through the process of setting up the Polar H10 and performing a sit-stand study, evaluating whether users can identify the correct actions, understand system feedback, and successfully complete the workflow.

2. Usability Benchmarking

Since this is a research-focused tool with few direct competitors, we will establish benchmarks based on stakeholder requirements and expert-defined protocols. We will measure whether the app meets specific performance standards (e.g., time to complete a sit-stand recording) as defined by our industry partner and primary research goals.

3. System Usability Scale (SUS) / SEOS

To quantify the user experience, we will implement a standardized scale (such as SUS or SEOS). This will allow us to evaluate the perceived ease of use and ensure that, despite being a technical research tool, the interface remains accessible and convenient for the end user.

4. Internal Agentic Testing (Unique Internal Method)

For our internal testing phase, we will utilize Agentic Testing. We will develop AI-driven agents assigned specific personas that mirror our target user group. By running these personas through the app repeatedly, we can identify edge cases, flow bottlenecks, and navigation errors before moving to human subjects.

Rationale & Feasibility

We chose Usability Benchmarking because the lack of direct competitors makes traditional competitive analysis difficult; instead, we need to ensure the app functions correctly according to the stakeholders who will utilize the data.

The System Usability Scale was chosen to provide a "check and balance" on the research-heavy features, ensuring they don't compromise the user experience. Overall, this plan is the most feasible approach within our time constraints, allowing us to leverage previous connections for recruitment while using agentic testing to maximize our internal QA process.

Participants

Participants consisted of individuals from our target demographic: women who are currently pregnant or have recently been pregnant, as this group represents the primary users of the TruePulse Health system. Due to limited availability within this population, recruitment focused on accessible participants through personal networks and prior study contacts.

Additional internal testing was conducted by team members and AI-based simulated users (agentic testing) to evaluate system behavior and edge cases prior to broader user testing.

Recruitment Strategy

We aimed to recruit 5+ users per test using the following methods:

- Personal Networks: Reaching out to individuals within our immediate social and professional circles who meet the pregnancy criteria.
- Longitudinal Outreach: Contacting previous participants who provided insights during our initial interview phase.
- Snowball Sampling: Asking current contacts to refer others in the target demographic.

User Availability: Low. Our target demographic consists of women currently experiencing pregnancy or who have recently been pregnant, making their time and physical availability highly constrained.

Protocols

For each type of usability testing, you must describe the protocols:

- Protocol for usability test

Objective

To evaluate whether users can successfully complete core tasks—specifically **setting up the Polar H10 sensor and completing the sit-stand protocol**—while ensuring the system is intuitive, efficient, and provides clear feedback.

Methodology

We used a combination of:

- **Cognitive Walkthrough (internal evaluation)** to assess learnability and task flow
- **Usability Benchmarking** to measure performance against stakeholder-defined standards
- **SUS/SEOS surveys** to capture perceived usability

Testing was **moderated** and conducted in a **hybrid format (remote and in-person)** depending on participant availability.

Tasks

Participants were asked to complete the following core workflow:

- Set up and wear the Polar H10 sensor
- Complete initial onboarding and survey
- Perform the sit-stand protocol sequence
- View and interpret results

These tasks reflect the app's three primary features: **profile setup, protocol recording, and results/history viewing**.

Participant Profiles Tested

- Primary users: Pregnant or recently pregnant individuals
- Secondary/internal testers: Team members simulating first-time users

- Agentic testers: AI-driven personas representing target users to explore edge cases and usability issues

Venue

Testing was conducted in a **hybrid environment**, including:

- Remote sessions (to accommodate participant constraints)
- In-person sessions when feasible

All sessions were moderated by one or more team members.

Usability Testing Results

1. Moderated Think Aloud

Method: Moderated Think-Aloud Usability Test

A moderated think-aloud usability test is a qualitative research method in which participants verbalize their thoughts while interacting with a product or prototype, in the presence of a trained moderator. According to Nielsen Norman Group, think-aloud testing is one of the most valuable usability evaluation methods because it reveals the reasoning behind user actions, not just the actions themselves — exposing confusion, incorrect mental models, and gaps in instructional clarity in real time.

This method was selected for TruePulse's first evaluation round because the application involves a novel, medically adjacent workflow (ECG sensor setup and an ANS monitoring protocol) that participants are unlikely to have encountered before. Understanding where users misinterpret instructions, feel uncertain about data, or lose trust in the system required direct, observable verbal feedback rather than quantitative metrics alone. The think-aloud protocol allowed the team to capture nuanced insights such as confusion around clinical terminology, uncertainty about correct behavior during the protocol, and difficulty interpreting numeric HRV outputs.

2. Test Documentation

2.1 Objective

The primary objective of this usability test was to evaluate the ease of use, clarity of instructions, and interpretability of results across the three core flows of the TruePulse mobile application:

- Sensor setup and onboarding (SetupInstructions.tsx)
- The active sit-stand ANS monitoring protocol (Protocol.tsx)
- Post-session results and history review (HistoryDetail.tsx / Journal.tsx)

This test was appropriate for the context because TruePulse targets postpartum users — a population with diverse technical literacy, heightened health awareness, and specific physiological concerns including brain fog and HRV recovery. Early-stage moderated testing with representative participants allowed the team to surface critical usability failures before development investment deepened, particularly around medically framed language and data literacy.

2.2 Tasks

Participants were given a single, open-ended task designed to mirror real-world use. No step-by-step guidance was provided in order to preserve ecological validity:

Primary Task (Read to Participant)

- Task: "Whenever you're ready, please begin by setting up the system and starting the session."
- Sub-task 1: Set up the wearable ECG sensor using the in-app instructions.
- Sub-task 2: Complete the guided sit-stand ANS monitoring protocol.
- Sub-task 3: Review your results at the end of the session.

Moderators used non-leading prompts if participants went silent: "What are you thinking right now?", "What do you expect to happen next?", and "What would you do if I wasn't here?" Participants were not corrected or redirected unless fully blocked.

2.3 Participant Profiles

Five participants completed the usability test. All were selected to reflect the target user group: individuals with recent or ongoing perinatal experience, representing a range of prior health technology familiarity.

- Participant 1 was a female with no prior experience using health apps and low contextual knowledge of ANS monitoring. She approached the session as a complete first-time user with no established expectations of the domain.
- Participant 2 was a female with prior health app experience and high contextual knowledge. She navigated the application with confidence and encountered minimal friction throughout all three task phases.
- Participant 3 was a female with prior health app experience but low tolerance for clinical or technical language. She had the most critical feedback regarding language clarity, step sequencing, and results readability.
- Participant 4 was a female with no prior health app experience and low contextual knowledge of physiological monitoring. She was engaged and curious throughout but expressed a strong need for interpretive guidance on results.
- Participant 5 was a female with prior health app experience and moderate contextual knowledge. She was sensitive to instructional gaps and safety guidance, particularly around physical transitions during the protocol.

2.4 Venue

Sessions were conducted in a remote, moderated format. Participants accessed the TruePulse prototype from their personal mobile devices, with the moderator present via video call. Session notes were recorded in real time by the moderator. Post-task questions were asked verbally before participants completed the digital post-test survey. This format was chosen to reflect the real-world, home-based context in which TruePulse is intended to be used, and to minimize travel burden for participants in the postpartum period.

3. Findings

3.1 Summary of Results

The five participants produced an average overall usability rating of 8.2 out of 10 on the post-test scale. SUS-equivalent scores ranged from approximately 41 to 96, indicating high variance between experienced and first-time users. The composite data across all participants is summarized in the table below:

Participant	Prior App Use	Rating	SUS Est.	Most Confusing	Most Helpful	Change Request
P1	No	9/10	~84	Sit/stand notifications unclear	Heart rate tracking visible	Improve sit/stand notifications

P2	Yes	10/10	~96	Nothing	Average heart rate display	Nothing to change
P3	Yes	4/10	~41	Steps/time unclear; results hard to read	Step sequence; strap fit	Friendlier language (like Flo app)
P4	No	9/10	~84	Results explanation missing	Heart rate in upper left	Sound when timer ends
P5	Yes	9/10	~84	Reading results at end	Sensor setup visuals	More instructions/warnings

Note: SUS scores above are estimated from the 10-point overall rating using a proportional conversion for summary purposes. A formal multi-item SUS score was not separately computed in this evaluation round.

Across the app-specific questions covering sensor setup clarity, protocol understanding, and results interpretability, the sit-stand protocol and results screens scored lowest. Three of five participants indicated difficulty understanding what was happening during the session or what the output numbers meant. Sensor setup instructions received mixed scores — generally high for experienced users and notably lower for novice participants.

3.2 What Surprised Us

Several findings were unexpected and surfaced insights not anticipated during the design phase:

- The "behavioral void" during the active protocol phase was the most surprising finding. Participants were uncertain whether talking, adjusting their clothing, or looking at their phone would invalidate the ECG measurement. This anxiety may itself have elevated heart rate, potentially skewing the very baseline the app was attempting to capture.
- The clinical term "electrode" caused disproportionate confusion. Even participants with general technology familiarity were unfamiliar with this word in a consumer context. The impact of a single vocabulary choice on perceived complexity and confidence was striking.
- Participant 2 (high familiarity) rated every category 5/5 and found nothing to change. This confirmed the app performs well for experienced users but exposed a significant experience gap between novice and expert participants — a gap the design must bridge.
- The request for a "help" section to explain results was raised independently by multiple participants who had not communicated with each other. The unprompted convergence on this need underscores it as a genuine and urgent design gap.
- Participant 3's reference to the Flo app as a usability benchmark was significant. It signals that users arrive with expectations shaped by consumer-grade, plain-language health apps — not clinical interfaces — and the product must meet that bar.

3.3 Changes to the Prototype

Based on the findings above, the following changes will be implemented in the next prototype iteration:

Immediate Changes (High Priority)

Language & Onboarding

- Replace "electrode" with plain-language terms such as "black rubber pads" or "sensor contact points" throughout SetupInstructions.tsx.
- Add a Pro-Tip overlay to Step 1 of the Setup Wizard: "A damp cloth or spray bottle works best — the strap should feel cool and moist."
- Replace or supplement click-through setup screens with a 30-second instructional video, as strongly requested by Participant 1.
- Remove orphaned "dot" pagination UI elements where swiping is not supported, to eliminate misclick confusion.

Protocol Behavioral Coaching

- Add a "Stillness Reminder" panel to Protocol.tsx during the 180-second Sit phases, coaching users on acceptable behavior.
- Include a subtle icon set indicating: No Talking, Relaxed Hands / Palms Up, Slow Breathing.
- Add a visible progress bar displaying the full sit-stand interval timeline with time remaining, to reduce ambiguity about session length.

Results Interpretability

- Add a "What does this mean?" tooltip to the RMSSD (HRV) card in HistoryDetail.tsx.
- Introduce color-coded status zones (e.g., green / yellow / red) to provide an immediate at-a-glance health signal.
- Add a Help section or glossary accessible from the results screen to define Recovery Insight, RMSSD, and Peak HR for non-clinical users.
- Display a postpartum-specific healthy range benchmark alongside numeric outputs so users can contextualize their scores.

Future Enhancements (Lower Priority)

- Explore a Journal.tsx enhancement that explicitly surfaces correlations between HRV dips and reported cognitive fatigue or brain fog journal entries.
- Add an audible timer chime at protocol phase transitions, as independently requested by Participant 4.
- Add a safety advisory for physical transitions during the protocol (e.g., "Stand up slowly") per Participant 5's feedback.

Agentic Testing

Objective

The primary objective of this usability evaluation was to identify critical friction points, failure modes, and mental model mismatches within the TruePulse Health prototype — specifically across three core user flows:

- Symptom logging via the My Journal feature
- Product discovery and adding items to cart via Shop & Resources
- Interpreting personal health data via the Analytics dashboard

Agentic simulation was selected as appropriate for this stage for several reasons. The prototype was in early formative development, and the team needed rapid, iterative feedback before recruiting live participants. Running AI-simulated tests allowed the team to stress-test the prototype across a wide range of pregnancy personas, emotional states, trimester stages, and technical literacy levels simultaneously — coverage that would be costly and time-consuming with human participants at this phase. The method is well-suited for identifying structural and architectural failures (crashes, broken routes, data siloing) and mental model mismatches that do not require the nuance of lived human experience to surface.

Tasks

Each participant (agent) was given the following three tasks:

Task 1: Log a Symptom Using My Journal

User intent: Record a specific physical or emotional pregnancy symptom (varied by persona: nausea, pelvic pain, acid reflux, contractions, spotting, etc.) using the journal feature.

Success criteria — Functional: Symptom is saved and confirmed in the journal.

Success criteria — Perceived: User feels the symptom was successfully recorded and will be tracked.

Task 2: Add a Recommended Product to Cart

User intent: After logging, navigate to the Shop & Resources section and add a relevant product to the cart.

Success criteria — Functional: A product is found and added to the cart.

Success criteria — Perceived: User trusts the recommendation is relevant and safe for pregnancy.

Task 3: Interpret Personal Health Insights from Analytics

User intent: Navigate to the Analytics dashboard and interpret personal health data, looking for symptom trends over time.

Success criteria — Functional: User navigates Analytics and views the Symptoms Overview.

Success criteria — Perceived: User believes the data displayed reflects their own inputs.

Participant Profiles

A total of 20 simulated participants were tested across three evaluators (Dinesh: 2 tests, Fernando: 8 tests, Sahil: 10 tests). All personas represented pregnant women at varying stages, tech literacy levels, and emotional contexts — the core target user group for TruePulse Health. Below are representative profiles:

P1 (Maya Reyes)	29	26 wks — 2nd trimester	Moderate	Low — first pregnancy	Cautiously optimistic, mildly impatient
P2 (Sarah Bennett)	34	32 wks — 3rd trimester	High	High — second pregnancy	Overwhelmed, exhausted, very low patience
P3 (Sarah Jenkins)	~20s	16 wks — 1st trimester	Low–Moderate	Low — first pregnancy	Exhausted, easily overwhelmed
P4 (Chloe Davis)	~20s	8 wks — 1st trimester	Moderate	Low — history of early loss	High anxiety, hyper-vigilant

P5 (Jessica Tran)	~30s	22 wks — 2nd trimester	High (PM)	High	Motivated, highly impatient with friction
P6 (Maria Gonzalez)	~30s	38 wks — 3rd trimester	Low	Low — acute physical distress	Extremely anxious, physically exhausted
P7 (Olivia Brooks)	~20s	12 wks — 1st trimester	High (athletic)	Moderate	Energetic, easily distracted by bugs
P8 (Emily Chen)	~30s	24 wks — twin, GDM	Very High (clinical)	High	Analytical, demands precision
P9 (Isabella Costa)	~30s	30 wks — non-native speaker	Low	Low	Fatigued, relies on visual cues
P10 (Sarah — Sahil)	28	2nd trimester	Moderate	Low	Motivated, mildly anxious
P11 (Chloe — Sahil)	32	34 wks — 3rd trimester	Moderate	Moderate	Overwhelmed, seeking reassurance
P12 (Sarah — Sahil T0)	29	1st trimester	Moderate	Low — first pregnancy	Overwhelmed, physically drained, zero patience

Note: Additional Sahil participants (Tests 3–9) represent a further range of pregnancy stages and contextual backgrounds with broadly similar profiles to those above.

Venue

All tests were conducted remotely in a controlled, simulated environment. The prototype (TruePulse Health, built in Vite/React) was served locally at <https://localhost:5173> or <https://localhost:5174> on a development machine. Agentic participants accessed the prototype through a structured prompt environment that replicated a mobile browser interaction context. No physical lab space was used. Tests were asynchronous and conducted on April 21–22, 2026.

Findings

Summary of Results

Across all 20 sessions, the evaluation revealed four systemic failure categories that affected the majority of participants, regardless of persona or test version:

Custom Symptom Input Trap	Critical	~16/20	The two-step custom symptom input (type text → click small '+' button → click Save) was non-intuitive. A majority of participants typed their symptom and clicked Save directly, silently discarding the entry. The '+' button had no visible affordance and did not respond to the Enter key.
Shop & Resources UI Occlusion + Fatal Crash	Critical	~14/20	The floating bottom navigation bar physically covered the Shop & Resources card on mobile viewports, making it unclickable. When participants attempted to navigate directly to /shop via the URL bar, the app consistently crashed with a black screen.
Identity / State Persistence Failure	Critical	~12/20	After completing the onboarding/setup flow, the app reverted user profiles to a default state (name: 'Elena', goal: 'Postpartum'). Pregnant users

			were then shown postpartum-specific symptoms and content, creating an immediate and severe trust breakdown.
Custom Symptoms Not Reflected in Analytics	Critical	~18/20	The Analytics dashboard displayed hardcoded/mock data (e.g., Fatigue 12x, Anxiety 8x) and did not integrate any user-generated custom symptom logs. Users who successfully logged symptoms found them completely absent from their analytics view.
Symptom Library Mismatch	High	~15/20	The default symptom list was populated with mood/postpartum states (Anxiety, Depression, Forgetfulness) rather than common physical pregnancy symptoms (Nausea, Heartburn, Pelvic Pain, Swelling). This forced all users into the broken custom input flow even for common symptoms.
Missing Product Detail Pages	High	~10/20	Tapping product cards in the shop opened no detail page — no ingredients, no safety information, no description. For pregnant users evaluating OTC medications and supplements, this

			was described as a critical safety concern.
Context-Blind Product Recommendations	Medium	~12/20	Recommendations shown after logging were static and did not respond to the specific symptom logged. A user logging nausea was shown anxiety-management books; a user logging a blood sugar spike was shown generic wellness products.

Task completion rates by task:

- Task 1 (Log symptom): ~25% full success; ~60% partial success; ~15% complete failure (crash or abandon)
- Task 2 (Add to cart): ~30% success; ~70% failure (occlusion, crash, or wrong routing)
- Task 3 (Interpret analytics): ~10% perceived success; ~90% experienced data trust breakdown

What Surprised Us

Several findings were unexpected and warrant particular attention:

1. The severity of the state persistence failure

The magnitude of the identity erasure bug was surprising. We expected onboarding to be a friction point, but did not anticipate that all setup data (name, pregnancy stage, goals) would be silently overwritten upon reaching the home screen. Seeing a stranger's name and postpartum health content when expecting personalized pregnancy data produced the most extreme emotional reactions across all sessions — immediate and total loss of trust, with several participants expressing they would delete the app immediately.

2. The Shop crash was universal

We anticipated that navigation to the shop might be difficult, but we did not expect the /shop route to produce a fatal crash (black screen) for virtually every participant who attempted to navigate there directly. This means that one of the three primary task flows was functionally inaccessible to all participants — a severity level higher than anticipated at this prototype stage.

3. Users in distress cannot compensate for friction

Across multiple tests, participants simulating users in acute physical or emotional distress (contractions, spotting, severe nausea) showed dramatically lower tolerance for even minor interaction friction compared to calmer personas. A two-tap custom input that a calm, tech-savvy user might work around caused complete task abandonment for a user simulating contractions. This highlighted how critical low-friction, error-tolerant design is for a health application serving vulnerable users.

4. The 'Notes' field drew more attention than the custom symptom field

Multiple participants across different evaluators independently typed their symptom into the Notes text field rather than the smaller custom symptom input — suggesting a consistent visual hierarchy problem where the larger, more prominent Notes field is interpreted as the primary data entry point for symptoms.

5. Analytics trust collapse was near-universal

We expected some confusion around analytics, but the extent of the trust collapse was striking. The combination of hardcoded mock data and complete absence of user-submitted symptoms meant that no participant in the evaluation could verify their own data was being tracked. For a health tracking application, this represents a fundamental breakdown of the core value proposition.

Prototype Changes to Be Made

Based on the findings, the following changes will be prioritized for the next prototype iteration:

Custom Symptom Input	P0 — Critical	Replace the two-step add flow (type → '+' → Save) with a single-step entry: pressing Enter or tapping outside the field auto-adds the tag. Increase the hit area of the '+' button and add a clear keyboard affordance.
Shop Route Crash	P0 — Critical	Debug and fix the /shop route. Ensure the route is registered and the component renders without error. Add error boundaries

		to prevent black-screen crashes.
Nav Bar Occlusion	P0 — Critical	Add sufficient bottom padding to all scrollable content areas to ensure the floating nav bar does not overlap tappable content cards.
State Persistence Bug	P0 — Critical	Fix the onboarding state management so user-entered data (name, pregnancy stage, goals) persists to the home screen and throughout the session. Remove hardcoded 'Elena' / 'Postpartum' defaults.
Analytics — Custom Symptom Integration	P0 — Critical	Connect user-submitted custom symptom logs to the Analytics visualization layer. Remove hardcoded mock data fallbacks when real user data exists.
Symptom Library	P1 — High	Redesign the default symptom list to include common physical pregnancy symptoms (Nausea, Fatigue, Heartburn, Pelvic Pain, Swelling, Spotting, Headache, Back Pain) organized by trimester. Consider a tabbed toggle between Physical and Mood categories.
Product Detail Pages	P1 — High	Add a product detail modal or page for every shop item, surfacing ingredients, pregnancy safety certifications, and brief descriptions. This is a trust and safety requirement, not a nice-to-have.
Contextual Recommendations	P2 — Medium	Implement dynamic recommendation logic that maps logged symptoms to relevant product/resource categories, replacing static curated lists.

Analytics — Trend Visualization	P2 — Medium	Replace count-only bar charts with timeline/calendar views showing when symptoms occurred. Users need to see frequency trends over time, not just totals.
Visual Hierarchy — Notes vs. Custom Symptom	P2 — Medium	Visually differentiate the Notes field from the Custom Symptom input. Consider placing the custom symptom entry more prominently, or providing a clear label: 'Add symptoms here' vs. 'Optional notes'.
Onboarding Performance	P3 — Low	Investigate and fix unresponsive Continue buttons during setup flow. Remove artificial delays.

Final Prototype (key screens)

Changes Made After Usability Testing & Rationale

Based on the System Usability Scale (SUS) findings, several key usability gaps were identified, particularly around **interpretability of data, clarity of actions, and ease of input**. Participants—especially first-time or low-context users—struggled with understanding outputs, navigating flows, and efficiently logging information. In response, we implemented the following design changes:

1. Global Quick-Add Journal (+ Button)

Change Made: Introduced a floating “+” button accessible from any screen, allowing users to quickly log journal entries, including baby kicks, weight, and vitamins.

Why:

SUS results and qualitative feedback showed that users needed a **faster, lower-friction way to log symptoms and behaviors**, particularly in moments of discomfort or urgency. The original flow required navigating through multiple screens, which increased cognitive load and risk of abandonment.

Options Explored:

- Embedding logging within each individual screen
- Keeping logging limited to the Journal section
- Adding a global action button (chosen)

Why This Option:

The floating action button was selected because it **minimizes navigation effort and supports real-time data capture**, aligning with the need for quick symptom logging identified in both usability testing and functional requirements (e.g., “record a symptom episode quickly with minimal interaction”) .

2. Interactive Timeline & Edit Mode

Change Made: Enhanced the Journal timeline to display summarized data (e.g., “140 lbs”) and added the ability to view details and edit past entries.

Why:

Participants expressed difficulty **reviewing and interpreting past data**, as well as a lack of control over previously entered information. SUS findings highlighted gaps in **results of interpretability and user control**, particularly for novice users.

Options Explored:

- Static timeline with no editing
- Separate “edit history” section
- Inline editing within timeline (chosen)

Why This Option:

Inline editing within the timeline was chosen because it **supports recognition over recall** and gives users direct control in context, reducing friction and improving trust in the system.

3. Resource Recommendation Pop-up

Change Made: Added a contextual pop-up that suggests relevant resources when users log concerning symptoms (e.g., anxiety).

Why:

A major insight from testing was that users **felt uncertain about what their data meant and what actions to take**, especially when logging symptoms. Multiple participants requested more **guidance and reassurance** when interpreting health information.

Options Explored:

- Static resource library users navigate manually
- Embedding resources within analytics screens
- Contextual, trigger-based recommendations (chosen)

Why This Option:

Contextual pop-ups were selected because they provide **just-in-time guidance**, reducing cognitive load and supporting decision-making without requiring users to search for help.

4. Simplified Analytics (Bonus Change)

Change Made: Replaced complex, data-heavy graphs with a simplified summary card showing average heart rate and animated visual.

Why:

SUS results showed that **3 out of 5 participants struggled to interpret results**, and users frequently requested clearer explanations and simpler outputs. Dense graphs increased confusion and anxiety rather than insight.

Options Explored:

- Keeping full clinical graphs
- Adding explanations on top of graphs
- Replacing with simplified summaries (chosen)

Why This Option:

The simplified summary approach was chosen to **prioritize clarity over completeness**, aligning with the project's goal of reducing anxiety and making physiological data understandable for non-expert users.

Overall Design Direction

Across all features, the chosen solutions prioritize:

- **Reduced cognitive load**
- **Faster interaction flows**
- **Improved interpretability of health data**
- **Increased user control and confidence**

These changes directly address the usability gaps identified in SUS testing—particularly for novice users—while maintaining functionality for more experienced users, helping bridge the usability gap observed between these groups.

Good Morning, **Elena**



Start Protocol

9 minute sit-stand ritual

My Journal



Analytics



Shop & Resources >

Home



Mar 26 - Apr 01

My Journal

Record your day, track your symptoms, and keep your memories safe.



Symptoms



Baby Kicks



Weight



Prenatal Vitamin



Memories



Custom



View My Journal Timeline

2 e



RESOURCE

Finding a... pump

Qualify through your insurance and get

Mar 26 - Apr 01

My Journal

Baby's First Kicks

×

DATE & TIME

20/04/2026

11:25 PM

Did you know?

At the start of the third trimester, it's recommended to count your baby's kicks daily to monitor their health.

Timeframe

Last 11 hours

Number of Kicks

-

2

+

Notes

Save

Shop Resources

Curated essentials for your journey.

Books

Magazines

Pharmaceuticals

Over



PRENATAL MULTIVITAMIN

Nature's Blend

★★★★★ 4.7

\$19.99

ADDED



DHA OMEGA-3 SUPPORT

Nordic Pure

★★★★★ 4.8

\$24.99

ADDED



FOLIC ACID 800MCG

Pure Pharma

★★★★★ 4.6

\$12.99

ADD



IRON HIGH POTENCY

Vita Health

★★★★★ 4.5

\$14.49

ADD

+

⌂

🕒

👤

My Analytics

Your health insights and trends.



Average Heart Rate

Recorded across breathing sessions

69 BPM



Symptoms Overview

Based on your journal entries



SESSIONS

2

ENTRIES

2



References

Unless otherwise required by your industry partner, please use APA or IEEE style for references
[single spaced with double spaces between entries]

Appendices

What might be included in the appendices:

- Project plan with dates
- Text of email sent to potential user research participants
- Survey
- Interview protocol
- Experience sampling guidelines for potential research participants
- Visual representations of qualitative user research data
- Any scales or instruments you used in Usability testing (likert scales, rancho los amigos scale, etc)
- Link to previous semester final report
- System Usability Scale (SUS) Questionnaire
- Snapshots from ideation sessions (if you think they would be relevant to the project—if you used a novel approach)
- Other

Pregnancy Monitoring Interview Summary (PW Participants)

A. Warm-up & Background

Question	PW 1	PW 2	PW 3	PW 4	PW 5
How far along are you / experience so far?	~25 weeks pregnant, first pregnancy; overwhelming due to age (39), frequent appointments and tests; physically manageable	Postpartum; baby born Oct 15; pregnancy managed with diabetes	Postpartum; pregnancy included gestational diabetes and preeclampsia at 36 weeks	Postpartum (~9 months); home birth with midwife; BP issues late pregnancy and postpartum	29 weeks pregnant; first pregnancy; first trimester difficult due to nausea; currently peaceful and going well
Sources relied on for pregnancy health info	OB/GYN, printed pamphlets, PDFs, pregnancy books, audiobooks	Gynecologist, dietician, diabetes specialist, Google	Medicaid nurse calls, OB office, health department resources, family and friends, pregnancy book	Midwife, early hospital visits, pregnancy app community forums	Doctors in family (husband's friends, brother-in-law), ChatGPT for small questions, Google for very minor queries

Question	PW 1	PW 2	PW 3	PW 4	PW 5
What matters most when monitoring health	Baby's health, nutrition, stress reduction, avoiding gestational diabetes	Keeping diabetes under control and following dietary restrictions	Blood pressure control, blood sugar, mental health, attending appointments	Blood pressure, swelling, baby position, infection awareness	Monitoring blood pressure fluctuations; overall pregnancy stability

B. Understanding of Symptoms & Body Awareness

Question	PW 1	PW 2	PW 3	PW 4	PW 5
How do you track changes in your body?	Monthly BP and weight checks at doctor; tracking fetal movements; lifestyle changes	Blood sugar checks four times daily using a written chart; BP checked during sonography	Home BP cuff, Dexcom glucose monitor, written notes/diary	Home BP cuff late pregnancy and postpartum; mental tracking; midwife check-ins	BP and heart rate checked during routine doctor visits; no regular home tracking unless dizziness occurs
Symptoms paid most attention to	Bleeding (type and color), fetal movement	Blood sugar levels, food reactions	Dizziness, high BP, heart rate changes, protein in urine	Swelling in hands/feet, high blood pressure	Low BP symptoms (dizziness, balance issues); later concern as BP increased
Information about warning signs & confidence	Doctors explained what is normal vs concerning; feels confident	Learned gradually through frequent visits and reassurance	Told to watch for cramping, bleeding, BP changes; still hard to judge normal vs abnormal	Midwifery care emphasized body awareness; prior pregnancy experience helped	Confident recognizing low BP through physical symptoms; learned through experience rather than formal instruction
Times reassurance was needed	Bleeding episodes caused fear; reassurance from doctors	Loose motions episode required urgent	High BP at work led to hospitalization; postpartum	Early ultrasound anxiety due to alcohol exposure	Concern about lack of cravings, minimal weight gain, and small baby bump;

Question	PW 1	PW 2	PW 3	PW 4	PW 5
	and books helped	doctor consultation	clotting explained by doctor	before knowing pregnancy	wanted clearer guidance

C. Current Tools & Monitoring Experience

Question	PW 1	PW 2	PW 3	PW 4	PW 5
Tools used to monitor health	No personal devices; monitoring done at clinic	Glucose meter with hospital-provided chart; no apps	Dexcom G7 continuous glucose monitor with app; home BP cuff	Pregnancy app (later stopped); home BP cuff	Occasional use of digital BP machine; no daily tools or apps
How health information is accessed	Direct explanations from doctor during visits	Sonography results explained immediately; reports via phone or hard copy	Patient portal app, Dexcom app, phone calls	Midwife communication; pregnancy app for general expectations	Doctor visits, BP monitor readings, online searches, family doctors
How well tools meet needs / gaps	Relies on providers; no major gaps noted	Structured sugar tracking met needs	Seeing results before provider caused anxiety; data overwhelming initially	Apps increased anxiety when expectations didn't match experience	Distrust in accuracy of digital BP monitor; otherwise no strong need for tools

D. Anxiety, Safety & Communication Needs

Question	PW 1	PW 2	PW 3	PW 4	PW 5
Challenging situations requiring support	Bleeding scares; major lifestyle changes to reduce risk	Managing long-term diabetes during pregnancy	Preeclampsia diagnosis, hospitalization, postpartum complications	Kidney infection mistaken for labor; poor early hospital communication	Anxiety caused by BP shifting from low to high; questioned diet and lifestyle
How decisions are made when	Follow doctor instructions	Always consult doctor before acting	BP readings and symptoms guide	Check BP and consult midwife	Checks ChatGPT first; if unsure,

Question	PW 1	PW 2	PW 3	PW 4	PW 5
symptoms appear	(e.g., bleeding → hospital)		contacting provider		consults family doctor
How providers communicate information	Supportive, explanatory, reassuring	Frequent and reassuring communication	Clear explanations reduced anxiety	Midwife communication empowering; hospital care felt dismissive	Feels providers give minimal explanations and dismiss concerns as “normal”
Preferred format for education	Audio books and verbal explanations	Verbal instructions	Verbal explanations with summaries	Direct conversations; minimal technical detail	Interactive app with Q&A, shared experiences, and expert responses

E. Technology Expectations & Feedback

Question	PW 1	PW 2	PW 3	PW 4	PW 5
Most useful info in an app/wearable	Baby movement tracking, reassurance, risk indicators	Blood sugar trends	Blood pressure, glucose, explanations of results	Blood pressure and infection indicators	Blood pressure trends with explanations
Preferred alert style	Calm, clear guidance (not panic-inducing)	Doctor-mediated feedback	Alerts with explanation and context	Minimal alerts with human confirmation	Non-alarming alerts with clear explanations
Detail vs summary preference	Combination	Detailed	Combination	Summaries	Combination (summary first, details on demand)
Feelings about AI alerts	Open if accurate and supportive	Not addressed	Open if transparent and explained	Skeptical due to past anxiety with apps	Uses AI for reassurance; open if reasoning is clearly explained

F. Wrap-up

Question	PW 1	PW 2	PW 3	PW 4	PW 5
What would you change or enhance?	Less anxiety; clearer reassurance between visits	Nothing major; care worked well	Better explanations before seeing results	Better early-care communication	Clearer explanations of bodily changes and individual differences
Anything else that would help?	More reassurance tools between visits	No additional needs identified	Provider explanations alongside data	Respectful and empathetic care access	Better educational support for first-time mothers

Clinician Interview Summary

Section A: Warm-up & Context

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
Current role & experience	Professor of practice (informatics), ~5–6 years; prior healthcare data analytics leadership	Hospice nurse; ~15 years across pediatrics, triage, clinic, hospice	Obstetrician; ~5 years across health posts, centers, hospitals	Family nurse practitioner since 2012; prior pediatric RN	Certified anesthesiologist assistant; ~11 years, OR + OB	OB clinician involved in hypertensive management
Devices/systems used to monitor vitals	Wearables (e.g., watches), implantables, fetal imaging, BP devices	Manual and wrist BP cuffs, pulse oximeter, home/hospice equipment	Sphygmomanometer, pulse oximeter, thermometer, fetal Doppler	Manual BP cuff, pulse oximeter documented in EHR	OR monitors (EKG, BP, SpO ₂ , CO ₂), EMR	Bedside BP cuffs, hospital monitors

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
High-risk populations & key data	Familiar with maternal/neonatal cases; BP most critical	Limited professional exposure; personal prenatal experience	Preeclampsia/HTN: BP primary; symptoms important	Limited OB exposure; BP and HR most important	Preeclampsia/HTN: BP plus continuous monitoring and labs	Preeclampsia thresholds drive escalation

Section B: Workflow & Interpretation

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
How physiological data is checked/interpreted	Outpatient vitals → tests → escalation; inpatient continuous monitoring	Reviews equipment but relies heavily on patient presentation	Vitals combined with patient interview and fetal monitoring	Compare vitals to norms, then assess patient appearance	Continuous real-time OR monitoring; waveform interpretation	BP checked at intervals; thresholds trigger protocols
Tools used & when	Imaging in clinic; continuous fetal monitoring inpatient	Minimal tech in hospice; basic tools	Home BP/Doppler vs hospital monitors	Outpatient EHR-based workflow	Bedside monitors; arterial lines when needed	Bedside monitors in hospital
How changes are communicated	Concern about alarm overload; need for	Alarms frequent but unreliable; trusts	Patient self-reports and ER visits	EHR documentation and	Monitor alarms, EMR, in-person	Nurse communication and

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
	automation	patient more		follow-up calls	communication	secure messaging
Useful metrics	BP, vitals, symptom patterns	Visual assessment over alarms	BP, fetal heart rate, symptoms	BP, HR, temperature	BP, EKG, SpO ₂ , labs	BP values and patient behavior
Preference for complex data display	Automated, filtered alerts with thresholds	Simple, trustworthy displays	Graphs and tracings with interpretation	EHR-integrated summaries	Adjustable alarm limits with integrated labs	Columns over time (BP trends)
Trends vs single values	Trends more meaningful	Patient trend plus appearance	Both, interpreted clinically	Trends difficult in outpatient care	Continuous trends, EMR snapshots	Standardized trend formats
Early distress indicators	BP changes and symptoms	Patient appearance, oxygenation	Headache, bleeding, BP spikes	Out-of-range vitals	BP, labs, monitoring escalation	BP, HR, visual assessment

Section C: Pain Points, Challenges & Gaps

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
Overall experience with tools	Data overload; lack of automation	Alarms unreliable; technology secondary	Equipment limited in rural settings	Manual BP collection time-consuming	Fragmented data across systems	Manual data entry burdens nurses

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
Technology affecting response	Excess data historically ignored	Alarm fatigue and desensitization	Limited access delays care	Manual vitals slow workflow	Searching EMR for labs delays response	Messaging helps but adds steps
Alerts & alarms	Must be meaningful or they're ignored	Constant false alarms	Mostly patient-driven alerts	Not addressed	Adjustable but poorly standardized	Loud but adjustable
What makes alerts useful	Custom thresholds and AI filtering	Fewer, more accurate alerts	Education and clear thresholds	EHR integration	Patient-specific limits	Highlight abnormalities only
Data across devices/interfaces	Major historical barrier	Fragmented and confusing	Access varies by facility	EHR works relatively well	Labs separate from monitors	Nurse relays data
Barriers during transitions	Staffing needed to monitor data	Home monitoring compliance issues	Access challenges in remote care	Outpatient limits	Finding data during care	Shift handoff complexity

Section D: Technology, Mobility & Communication

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
When mobile access is helpful	High-risk outpatient monitoring	Limited usefulness	Home monitoring for BP and fetal HR	Not critical in outpatient setting	Helpful if labs and vitals combined	Secure messaging on phone
Small-screen display needs	Filtered, actionable alerts	Simple, reliable numbers	Clear normal vs abnormal indicators	EHR pop-ups	Integrated vitals and labs	Trend summaries

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
Communicating data to patients	Education and symptom awareness	Patients struggle interpreting tech	Explain normal ranges clearly	Not addressed	Not addressed	Not addressed
Thoughts on AI/predictive systems	Necessary but must reduce noise	Skeptical unless trustworthy	Helpful if interpretable	Interested but cautious	Useful if transparent	Needs outcome evidence

Section E: Wrap-up

Question	Clinician 1	Clinician 2	Clinician 3	Clinician 4	Clinician 5	Clinician 6
One thing to improve	Intelligent automation	Alarm reliability	Equipment access and education	Faster vitals capture	Unified data view	Reduce manual data entry
Needs for continuous platform	Custom thresholds and AI triage	Simple, patient-centric design	Clear ranges and escalation rules	EHR integration	Vitals, labs, and alarms together	Hospital-like display
Who approves new technology	Institutional leadership	Organization administration	Health system leadership	Practice leadership	Hospital systems	Hospital committees

Researchers using current tool to collect physiological data

Question	Researcher	Answer
Could you briefly describe your role and what a typical day looks like for you?	Researcher 1	Clinical research coordinator responsible for subject recruitment, scheduling, running study sessions, IRB protocol documentation, and acting as the liaison between the PI and the IRB. Typical days vary: sometimes running

Question	Researcher	Answer
		participants all day, other days focused on IRB paperwork, ordering lab supplies, or assisting with grant submissions.
	Researcher 2	Laboratory research specialist responsible for managing lab equipment and sensors used in experiments. Also runs participants, helps supervise undergraduate and graduate students, and occasionally writes Python scripts for data processing and cleaning.
What types of clinical studies are you currently running?	Researcher 1	Most studies are student-led projects connected to theses or training projects. Typical studies involve physiological monitoring experiments and behavioral tasks.
	Researcher 2	Some large workforce-development studies have been conducted with over 100 participants, but many smaller studies involve around 20–30 subjects.
Roughly how many participants are you responsible for monitoring at any given time?	Researcher 1	Usually one large study with around 100–200 participants and several smaller studies with about 20–30 participants each.
	Researcher 2	
Who else do you collaborate with when reviewing participant data?	Researcher 1	Collaboration typically involves the principal investigator (Greg), graduate students, and sometimes undergraduate researchers who help with parts of the data pipeline.
	Researcher 2	
Walk me through your process when you sit down to check participants' data. Where do you start?	Researcher 1	Data collection staff are trained to recognize what “good” physiological signals should look like. Monitoring often involves visually checking the signal feed and ensuring sensors and signals appear correct during data collection.
	Researcher 2	
What specific pieces of information are you looking for first?	Researcher 1	Focus on signal quality in physiological data (e.g., ECG signals) to ensure the sensor is functioning and producing valid readings.

Question	Researcher	Answer
	Researcher 2	
Tell me about the last time you noticed something concerning or anomalous in a participant's data. How did you spot it, and what did you do next?	Researcher 1	If the ECG signal looks abnormal, it may indicate a sensor problem (e.g., sensor fell off). Researchers pause the study session, adjust equipment, or restart depending on how far the session has progressed.
	Researcher 2	
Along with physiological data, do you collect other qualitative or quantitative data?	Researcher 1	Yes. Demographic surveys are always collected. Questionnaires such as the Body Perception Questionnaire are frequently used to measure participants' awareness and responses to bodily signals and other psychological or physiological factors.
	Researcher 2	
Currently, how do you review physiological data alongside self-reported survey data?	Researcher 1	Data streams are aligned based on timing. Physiological data (e.g., ECG) and task data are synchronized so everything exists on the same time scale before analysis.
	Researcher 2	
What tools or systems are you using right now to make sense of these different data types?	Researcher 1	Data processing and analysis tools vary by researcher but commonly include Python, R, and MATLAB. Data synchronization and analysis scripts are used depending on the analyst's background.
	Researcher 2	
Can you describe a time when it was difficult to correlate physiological metrics with survey data?	Researcher 1	

Question	Researcher	Answer
	Researcher 2	
When you see a sudden drop or spike in a metric, how do you determine why it happened?	Researcher 1	Researchers review live signals and check whether equipment or sensors malfunctioned. Observational notes from the session help explain anomalies (e.g., participant movement or interruptions).
	Researcher 2	
How do you document or share findings regarding a participant's status with the rest of your research team?	Researcher 1	During sessions, notes are written on a subject checklist. These notes are later compiled into a spreadsheet containing subject number, date, and observations so analysts reviewing the dataset can interpret anomalies.
	Researcher 2	
What is the most frustrating, time-consuming, or manual part of collecting or monitoring participant data?	Researcher 1	Recruiting participants and getting them to show up consistently is the biggest challenge due to high attrition between interest, scheduling, and actual participation.
	Researcher 2	Manual data cleaning after collection can be time-consuming. Automating artifact detection and cleaning would significantly reduce effort.
Is there anything else about monitoring study participants that we haven't discussed?	Researcher 1	Maintaining high-quality physiological signals depends heavily on participant cooperation and correct sensor placement. Poor data often stems from user behavior rather than system issues.
	Researcher 2	

Persona Drafts



Sarah Jenkins
Acute Care Clinician

About

 38

 12 hour shifts

 10+ Yrs Exp.

 High-Noise Unit

Skeptical

Pragmatic

Patient-Focused

Time-Poor

Empathetic

Tech-Cautious

Motivations

Sarah is driven by patient safety above all else. She views technology purely as a tool to serve the clinical goal, not an end in itself. Her primary motivation is to efficiently filter out environmental and data "noise" so she can focus her limited time on true clinical crises and human connection at the bedside.

Goals

- Assess immediate patient status (Stable vs. Unstable) in under 5 seconds upon entering a room.
- Reduce time spent deciphering "black and blue" EMR data to increase time for patient interaction.
- Explain complex physiological data to anxious families in plain language without causing panic.

Pain Points

- Alarm Fatigue:** She has become desensitized to constant monitor beeping because most alerts are caused by patient movement rather than real danger.
- The "Black Box":** EMRs provide automated risk scores without showing the underlying math, forcing her to trust data she cannot verify.
- Cognitive Overload:** Having to perform mental math for critical metrics (like Preeclampsia ratios) while under pressure at the bedside.

Tools & Technologies

- Manual Stethoscope/Cuff:** Her absolute "source of truth" for validated data when digital tools seem off.
- AI Predictions (Muse):** Used only as a loose guide for prioritizing patient lists, but rarely trusted blindly without human proofreading.
- Bedside Monitors:** Used for general trend awareness (up or down arrows), but specific snapshot numbers are often ignored due to noise artifacts.

Behaviors & Workarounds

- The Validation Loop:** If a digital cuff gives a critical reading, she immediately discards the data point and re-takes it manually to ensure accuracy.
- Ignoring Protocol:** She delays logging data for anxious patients ("White Coat Syndrome"), waiting 10 minutes for them to calm down to get a "true" baseline reading.
- Physical Corroboration:** She prioritizes looking at the patient physically (e.g., skin color, breathing ease) over monitor data if the two conflict.



Elena Rossi
Expectant Mother (High Risk)

About

 32

 28 weeks pregnant

Anxious

Proactive

Data-Sensitive

Community-Reliant

Condition:

- Gestational Diabetes & Preeclampsia Watch

Motivations

Elena is driven by the health of her baby above all else. She wants to feel "empowered" and in control of her pregnancy journey but is easily overwhelmed by raw medical data. Her primary motivation is to translate abstract numbers (BP, blood sugar) into actionable peace of mind, seeking human reassurance over algorithmic predictions.

Goals

- Establish Baselines:** Quickly understand "what is normal" for her specific body versus what is a generic pregnancy statistic.
- Actionable Synthesis:** Receive immediate, practical advice (e.g., "drink water," "rest now") rather than just seeing a red warning number or a graph she can't interpret.
- Bridge the Gap:** Efficiently communicate daily symptoms to her provider without waiting for the next monthly appointment or feeling like a burden.

Pain Points

- Data Anxiety:** Viewing raw results (like blood sugar spikes or BP fluctuations) before a doctor explains them causes immediate stress and fear for the baby's safety.
- The "Normalcy" Void:** Medical providers often dismiss symptoms as "normal" without explaining the physiology, forcing her to self-educate through Google or family.
- Algorithmic Rigidity:** Pregnancy apps often fail because they rely on standard calculations (like standard LMP) that don't match her unique medical reality, leading her to abandon them.
- Alert Fatigue:** She becomes stressed or annoyed by loud, intrusive auditory alarms for non-emergencies; she prefers subtle nudges.

Tools & Technologies

- Home Monitoring Devices:** She relies on a BP cuff and Glucose monitor as her daily "source of truth," though she sometimes doubts their calibration or accuracy.
- AI as "First Responder":** She uses tools like ChatGPT for immediate triage of low-stakes questions ("Is this symptom normal?") before escalating to a human.
- Community Forums:** Used to validate her physical sensations against the shared experiences of other mothers, which she trusts more than generic app articles.

Behaviors & Workarounds

- The Validation Loop:** If a digital app gives a prediction that conflicts with how she feels physically, she prioritizes her somatic awareness (body sensation) and ignores the data.
- Hybrid Information Seeking:** She cross-references everything: first asking AI/Google, then verifying with a family member (often a medical proxy), and finally confirming with her doctor.
- Manual Synthesis:** She keeps physical notes or mental logs of symptoms to present to her doctor because she feels digital apps don't capture the full context of her condition.

Internal Evaluation

Cognitive Walkthrough Report (CWR)

CWR Number: TPH-01
Product Name: TruePulse Health – ANS Tracking App
System: New phone system
Task Name: Successfully setting up the Polar H10 & conducting a sit-stand study Date: March 28, 2026
Experimenters: Aiden Eichenour, Dinesh Bhathad, Sahil Deshpande, Fernando Luna

Task Description

User is fitted with the Polar H10 sensor and completes a sit-stand study using the application.

Task Action Sequence

Step	User Action	System Response
1	Pick up & setup the Polar H10	System recognizes sensor presence
2	Begin setup sequence	Visual tracking validation
3	Complete pre-study survey	Survey displayed
4	Begin sit-stand sequence	Study begins
5	Follow study sequence (3-1-1-1-3)	Guided protocol progression
6	Finish test	Test complete
7	View results	Results displayed

Interface / System Description

The application runs on a standard mobile device and integrates with the Polar H10 chest strap sensor. The strap is adjustable and requires slight moisture for accurate readings. The app includes a pre-study survey, guided sit-stand animation, and a standardized results screen.

Cognitive Walkthrough

Test 1

Step 1: Begin Setup Sequence

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: User successfully set up profile and addressed privacy concerns.

Step 2: Setup Polar H10

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	Animation effectively demonstrated setup
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: Setup fully guided through the app due to lack of physical device feedback.

Step 3: Begin Sit-Stand Sequence

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: Clear navigation available on home screen.

Step 4: Follow Study Sequence

Question	Result	Notes
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Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	Audio cues draw attention
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: Needs validation for inaccurate data scenarios.

Step 5: Finish Test & View Results

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: User completed entire flow successfully with no errors.

Test 2

Step 1: Begin Setup Sequence

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	Visual instructions provided
Will the user see that progress is being made?	Yes	

General Notes: User successfully set up profile and addressed privacy concerns.

Step 2: Setup Polar H10

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	

Will the user see that progress is being made?	Yes	
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General Notes: —

Step 3: Begin Sit-Stand Sequence

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	Task requires correct execution
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	Data animation provides reassurance
Will the user see that progress is being made?	Yes	

General Notes: —

Step 4: Follow Study Sequence

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: Needs validation for noisy data scenarios.

Step 5: Finish Test & View Results

Question	Result	Notes
Will the user try to achieve the right effect?	Yes	
Will the user know the correct action is available?	Yes	
Will the user know the action leads to the desired effect?	Yes	
Will the user see that progress is being made?	Yes	

General Notes: User completed entire flow successfully with no errors.

Potential Fixes for Identified Issues

1. Generate higher-quality animation assets
2. Develop a storyboard for the sit-stand sequence
3. Improve visual and audio cues to maintain engagement without distraction

End of Cognitive Walkthrough Report